

A compact HALEU reactor parked at JUNO: Standard-Model precision, sterile-neutrino wiggles, and axion-like particles

Internal report on notebooks 5–7 of the JUNO repository

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This report summarises what I have done on the physics case for a compact high-assay low-enriched-uranium (HALEU) research reactor of 10 MW_{th} parked about 50 m from the JUNO central detector. Three studies are carried out within one analysis framework, built on the measured Daya Bay antineutrino spectra, JUNO’s released detector response, and analytic Asimov profiling over joint Gaussian nuisance modes. First, elastic neutrino–electron scattering (E ν ES) at $\sim 3 \times 10^3$ events per day gives a $\sim 2\%$ determination of the low-energy weak mixing angle when analysed alone (where the measured flux shape is the limit), and $\sigma(\sin^2 \theta_W) = 1.1 \times 10^{-3}$ (0.5%) when the 6×10^4 per day inverse-beta-decay (IBD) sample is used to anchor the flux in situ, with 5×10^{-4} within reach of plausible transfer systematics. Second, for $\Delta m_{41}^2 \sim 0.1\text{--}10\text{ eV}^2$ the sterile-neutrino oscillation length is metres, so JUNO’s 33 m diameter turns the detector into a many-baseline experiment. A shape-only, vertex-binned wiggle search reaches $\sin^2 2\theta_{14} \approx 10^{-3}$ over $0.2\text{--}3\text{ eV}^2$, an order of magnitude beyond the reactor short-baseline programme, with no dependence on the reactor flux normalisation or shape and with the full 3+1 formula carried across standoffs of 50–1400 m. Third, the core’s $\sim 5 \times 10^{18}\text{ }\gamma/\text{s}$ produce axion-like particles (ALPs) through Primakoff and Compton-like conversion; with an 855 m^2 face and 22 m mean decay path, JUNO covers the “cosmological triangle” of the ALP–photon plane in full and reaches $g_{aee} \sim 10^{-8}$ at MeV masses with the background included. Every cross section, decay rate and flux ingredient is validated against independent implementations or analytic benchmarks, and the systematics budgets and their limitations are stated explicitly.

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I. COMMON FRAMEWORK

All three studies share one set of inputs and one statistical machinery, so I describe them once here.

a. Source and fuel evolution. A HALEU core of thermal power P with fission fractions $f_i(\beta)$ ($i = {}^{235}\text{U}, {}^{238}\text{U}, {}^{239}\text{Pu}, {}^{241}\text{Pu}$; $\sum_i f_i = 1$) at burnup position $\beta \in [0, 1]$ in its cycle has fission rate and antineutrino flux

$$R_f(\beta) = \frac{P}{\sum_i f_i(\beta) E_i}, \quad \frac{d\Phi_\nu}{dE_\nu}(E_\nu, \beta) = \frac{R_f(\beta)}{4\pi L^2} \sum_i f_i(\beta) S_i(E_\nu), \quad (1)$$

with the thermal energies per fission $E_i = (202.36, 205.99, 211.12, 214.26)$ MeV [1] and S_i the per-fission antineutrino spectra. For $P = 10$ MW this gives $R_f = 3.07 \times 10^{17} \text{ s}^{-1}$ at $\beta = 0.5$. The fresh composition is $f_{235} = 0.98$, $f_{238} = 0.02$, and plutonium grows in along the *Daya Bay* fuel-evolution trajectory [2]: with the measured ${}^{239}\text{Pu}$ and ${}^{241}\text{Pu}$ fission fractions $F_{239,241}(\beta_k)$ across Daya Bay's 20 burnup groups (β_k mapped linearly onto $[0, 1]$),

$$f_{239}(\beta) = F_{239}(\beta) - F_{239}(0), \quad f_{241}(\beta) = F_{241}(\beta) - F_{241}(0), \quad f_{238} = 0.02, \quad f_{235} = 1 - f_{238} - f_{239} - f_{241}, \quad (2)$$

that is, the LEU *increments* of the Pu fractions are applied on top of the fresh HALEU composition at fixed ^{238}U . This is deliberately conservative: a 2%- ^{238}U core breeds plutonium more slowly than the LEU cores the trajectory describes, so the true spectral drift is smaller than modelled. At the run-averaged $\beta = 0.5$ used throughout, $(f_{235}, f_{238}, f_{239}, f_{241}) = (0.901, 0.020, 0.063, 0.016)$; at end of cycle $(0.818, 0.020, 0.125, 0.037)$. The fuel-evolution *systematic* is a single nuisance mode scaling the Pu increments by $(1 + \epsilon)$, $\sigma_\epsilon = 0.30$, whose spectral direction is

$$\frac{\partial}{\partial \epsilon} \sum_i f_i S_i = f_{239} (S_{239} - S_{235}) + f_{241} (S_{241} - S_{235}), \quad (3)$$

that is, fission share moved from ^{235}U into plutonium at fixed ^{238}U .

The per-fission spectra are the *measured* Daya Bay unfolded per-fission IBD yields for ^{235}U and ^{239}Pu [3], divided by the Vogel–Beacom cross section [4] wherever a flux rather than a yield is needed, together with the Huber–Mueller spectra [5, 6] for the percent-level ^{238}U and ^{241}Pu components (Section A). This flux is defined only above the IBD threshold of 1.806 MeV, so recoil spectra below $T \sim 1.5$ MeV are conservative.

b. Detector. I take JUNO’s fiducial target to be $N_p = 1.44 \times 10^{33}$ free protons [7, 8] and $N_e = 4.67 N_p = 6.7 \times 10^{33}$ electrons (LAB is $\sim 88\%$ carbon and 12% hydrogen by mass), inside a fiducial radius of 16.5 m. The response uses the released positron and electron non-linearities, an energy resolution $\sigma_E/E = \sqrt{a^2/E + b^2}$ with $a = 3.3\%$ and $b = 1.0\%$, a vertex resolution of $10 \text{ cm}/\sqrt{E}$, a selection efficiency of 0.8, and a reactor duty cycle of 0.9.

c. Statistics. All sensitivities are Asimov projections [9] with the covariance $C = \text{diag}(\mu) + \sum_k v_k v_k^T$, one unit-Gaussian mode v_k per systematic. Fisher matrices and $\Delta\chi^2$ then follow analytically, no nuisance minimisation is ever performed, and the Woodbury identity keeps fine binnings cheap. Any prediction that is affine in a parameter (the couplings, or $\sin^2 2\theta_{14}$) is scanned by templates.

A. When does the near reactor pile up?

A reactor at tens of metres produces IBD at roughly 1 Hz per 10 MW ($R_{\text{IBD}} = 0.69 \text{ Hz}$ at 10 MW and 50 m, scaling as P/D^2), each event being a prompt hit followed by a delayed neutron capture about $200 \mu\text{s}$ later. Three distinct pile-up questions arise, and they have very different thresholds.

(i) Accidental IBD-like coincidences. JUNO’s radioactivity singles above ~ 0.7 MeV run at $\sim 10^4 \text{ Hz}$ [8], and the reactor adds delayed neutron captures at R_{IBD} . Random pairings of a

radioactivity single with a reactor neutron inside JUNO’s coincidence window and 1.5 m distance cut are accidental IBD candidates. Both the accidental rate and the true IBD rate scale linearly with the reactor, so their *ratio* is a constant set by JUNO’s own singles. Scaling JUNO’s measured accidental-to-IBD ratio (0.9 versus 47 per day in the 59-day release [7]) gives about 2% accidentals at every power and distance (1.1×10^3 against 6.0×10^4 per day at 10 MW and 50 m). This is a background one *measures* from off-window sidebands, not a pile-up limit.

(ii) *Genuine IBD self-pile-up.* Two reactor IBDs within one coincidence window (~ 1 ms) confuse the prompt–delayed pairing. The probability is $1 - e^{-R_{\text{IBD}}\tau}$: 7×10^{-4} at 10 MW and 50 m, 0.7% at 100 MW and 50 m, 6.7% at 1 GW and 50 m. This is the operative boundary. Requiring less than 1% self-pile-up limits the source to

$$\frac{P}{D^2} \lesssim 0.06 \text{ MW m}^{-2}, \quad \text{i.e.} \quad P \lesssim 150 \text{ MW} \left(\frac{D}{50 \text{ m}} \right)^2, \quad (4)$$

which corresponds to $\approx 8.6 \times 10^5$ IBD per day whatever the geometry (145 MW at 50 m, 580 MW at 100 m, 2.3 GW at 200 m). Every configuration in this report (10 MW at 50 m, and up to 100 MW in the mobile-reactor study of notebook 2) sits comfortably inside it; a GW-class core would have to stand $\gtrsim 150$ m away.

(iii) *Single-hit dead time and DAQ.* The reactor’s total hit rate (prompt, delayed and $E\nu\text{ES}$) is 1.4 Hz at 10 MW and 50 m, and only 145 Hz even at 1 GW and 50 m, which is 0.01% to 1.5% of the radioactivity singles JUNO already triggers on. Event overlap within a $\sim 1 \mu\text{s}$ readout window stays at the $\sim 1\%$ level set by radioactivity, independent of the reactor. Single-hit physics is therefore never rate-limited by the reactor; its reactor-induced backgrounds are the *untagged* IBD positrons and the $E\nu\text{ES}$ events themselves, which are modelled explicitly in Sections II to IV. (The reactor’s own neutrons and gammas leaking through its shield are a separate, shielding-design question addressed in notebook 8; see Section V.)

The conclusion is that below the boundary of Eq. (4) pile-up is a percent-level nuisance measured in situ, and the physics reach is set by statistics and systematics, not by rate.

II. STANDARD-MODEL PRECISION WITH $E\nu\text{ES}$ (NOTEBOOK 5)

A. Cross sections and rates

The $E\nu\text{ES}$ cross section for $\bar{\nu}_\alpha e \rightarrow \bar{\nu}_\alpha e$ is

$$\frac{d\sigma}{dT} = \frac{2G_F^2 m_e}{\pi} \left[C_L^2 + C_R^2 \left(1 - \frac{T}{E} \right)^2 - C_L C_R \frac{m_e T}{E^2} \right], \quad C_{L,R} = \frac{1}{2}(C_V \mp C_A) \text{ (antineutrino)}, \quad (5)$$

with $C_V = g_V + \delta_{\alpha e}$, $C_A = g_A + \delta_{\alpha e}$ and the neutral-current couplings $g_V = 2 \sin^2 \theta_W - \frac{1}{2}$, $g_A = -\frac{1}{2}$ at tree level. The reference point used throughout is the PDG 2024 world average of the ν - e couplings, $g_V = -0.040$ and $g_A = -0.507$ [1], which coincides with the radiatively corrected SM prediction $(-0.0395, -0.5063$ [10]) and corresponds to an effective low-energy $\sin^2 \theta_W = (g_V + \frac{1}{2})/2 = 0.230$; sensitivities are quoted as errors on that effective $\sin^2 \theta_W$. Cross sections come from NEPTUNE [11], and the parametric (g_V, g_A) implementation used for the scans reproduces it to 3×10^{-16} at NEPTUNE’s own tree-level couplings ($\sin^2 \theta_W = 0.2223$, $g_A = -\frac{1}{2}$). Figure 1 compares the cross sections per LAB “molecule”, and Fig. 2 shows the recoil shapes at fixed E_ν . LAB is $\text{C}_6\text{H}_5\text{C}_n\text{H}_{2n+1}$ with $n \approx 12$, i.e. $\text{C}_{18}\text{H}_{30}$ or one $\text{CH}_{1.67}$ unit per carbon: one carbon nucleus, 1.67 free protons and 7.67 electrons, which is the stoichiometry behind $N_e = 4.67N_p$. (I also evaluated the trident channel $\bar{\nu}_e N \rightarrow \bar{\nu}_e e^+ e^- N$ with NEPTUNE, coherently on carbon and elastically on the free protons: at reactor energies it is $O(\alpha)$ and phase-space suppressed above its 1.02 MeV threshold and amounts to $\sim 10^{-5}$ of the $\text{E}\nu\text{ES}$ rate, 0.04 events per day at 10 MW and 50 m, so it is not carried further.) Four features drive the analysis. IBD is $\sim 100\times$ larger per molecule than $\bar{\nu}_e e$ even after the 7.67 electrons are counted, which is why the IBD sample is $20\times$ the $\text{E}\nu\text{ES}$ one. The antineutrino cross section lies *below* the neutrino one because the CC and NC amplitudes interfere destructively ($C_L = \frac{1}{2}(C_V - C_A)$ is small for $\bar{\nu}_e$). The NC-only $\bar{\nu}_{\mu,\tau}$ channel is a further factor ~ 3 down; it is irrelevant at 50 m but carried anyway. And the $T > 1$ MeV window removes about 60% of the $\bar{\nu}_e e$ rate at $E_\nu = 3$ MeV, less at higher energy. The $\bar{\nu}_e$ recoil spectrum is flat and then falls in T (the $C_R^2(1 - T/E)^2$ term), and it is the E_ν -dependence of that fall-off, not the total rate, that carries the $\sin^2 \theta_W$ information once the normalisation is free.

The flavour composition is exact: the $\bar{\nu}_{\mu,\tau}$ component from $1 - P_{ee}$ (6×10^{-4} at 50 m) is carried with its NC-only cross section. An electron only participates for T above its binding energy (carbon K 284 eV, L 19/11 eV, hydrogen 13.6 eV); this atomic-binding correction [12] is applied to every spectrum and is exactly unity above 1 keV, so it is completely negligible for the analysis window $T \in [1, 6.5]$ MeV.

The resulting rates for 10 MW at 50 m are 5.95×10^4 IBD per day ($\sim 1700\times$ the far reactors) and 3.2×10^3 $\text{E}\nu\text{ES}$ per day in the analysis window, 99.99% of them $\bar{\nu}_e$. Figure 3 shows the recoil spectrum with the backgrounds I model: IBD singles (untagged-neutron fraction 1%, 577 per day, measurable in situ from the 99%-tagged sample), solar neutrinos (^8B , hep and CNO continua plus the pep line [13, 14], LMA flavour-weighted, 182 per day live-equivalent). Radioactivity and cosmogenic singles are *not* modelled and would dominate below ~ 3 MeV, so the precisions below are reactor-limited ones.

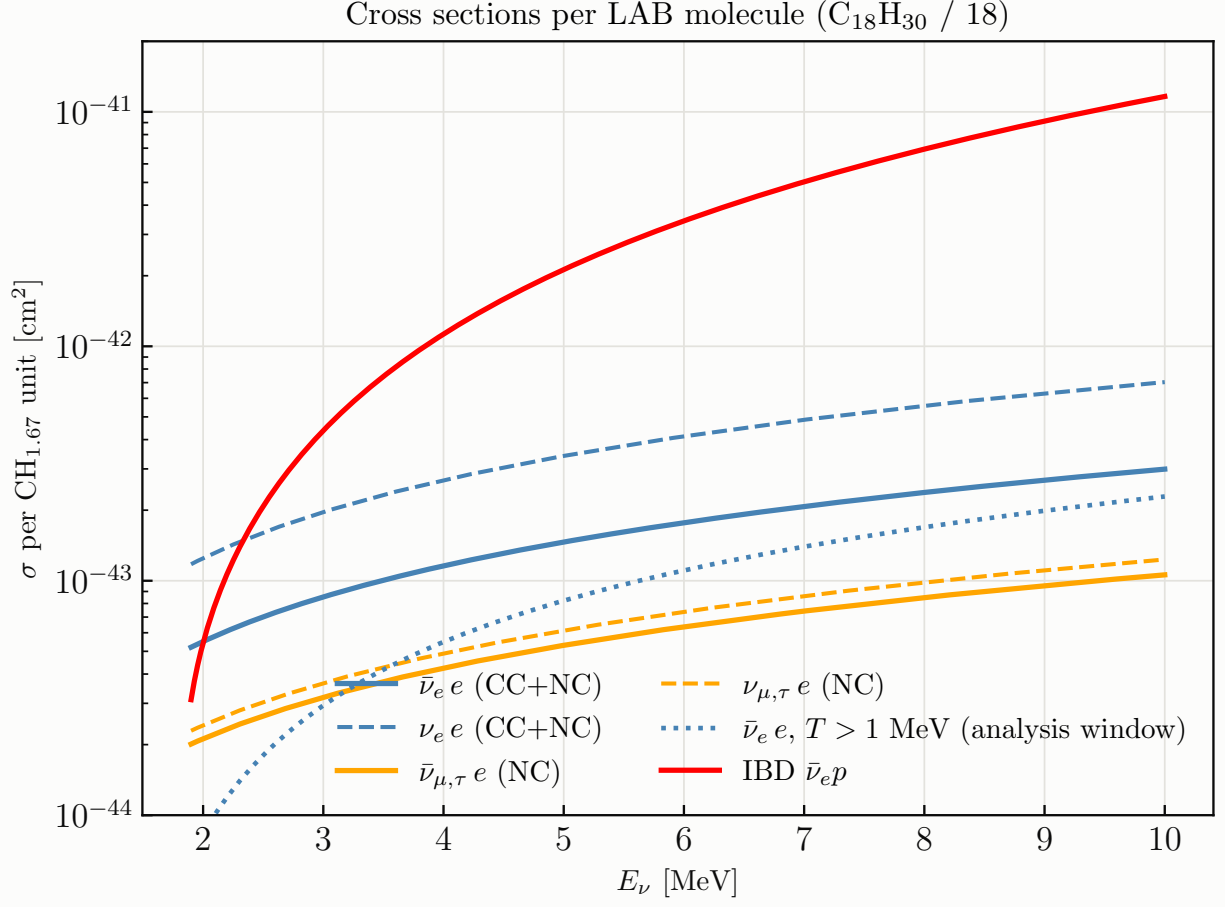


FIG. 1: Cross sections per LAB molecule (one $\text{CH}_{1.67}$ unit: 1 carbon, 1.67 free protons, 7.67 electrons): $\text{E}\nu\text{ES}$ for $\bar{\nu}_e/\nu_e$ (CC+NC) and $\bar{\nu}_{\mu,\tau}/\nu_{\mu,\tau}$ (NC), the $\bar{\nu}_e e$ rate restricted to the $T > 1$ MeV analysis window, and IBD on the free protons [4].

B. Systematics

The $\text{E}\nu\text{ES}$ -only fit carries the following modes: a 2% normalisation (power $\times$ target $\times$ efficiency); the 25-mode measured ^{235}U shape covariance [3]; ^{238}U at $\pm 15\%$; the IBD-singles normalisation at $\pm 10\%$; solar at $\pm 3\%$; energy scale, bias and resolution at 0.5/0.5/5%, coherent between the electron and positron responses; and the fuel-evolution (Pu-ingrowth) mode at $\pm 30\%$.

The joint fit stacks the IBD anchor spectrum (1.0–9.0 MeV in 50 keV bins) with the $\text{E}\nu\text{ES}$ spectrum, *frees the flux normalisation* (a loose 10% prior) and lets the IBD channel measure it. The transfer to $\text{E}\nu\text{ES}$ is then limited by the IBD cross-section normalisation (0.2%, from τ_n and radiative corrections [4, 15]), a zero-mean linear cross-section tilt (0.1%), and the channel ratio $N_e/N_p \times \varepsilon$ (0.5%). Under the anchor the standard JUNO backgrounds (far reactors, geoneutrinos,

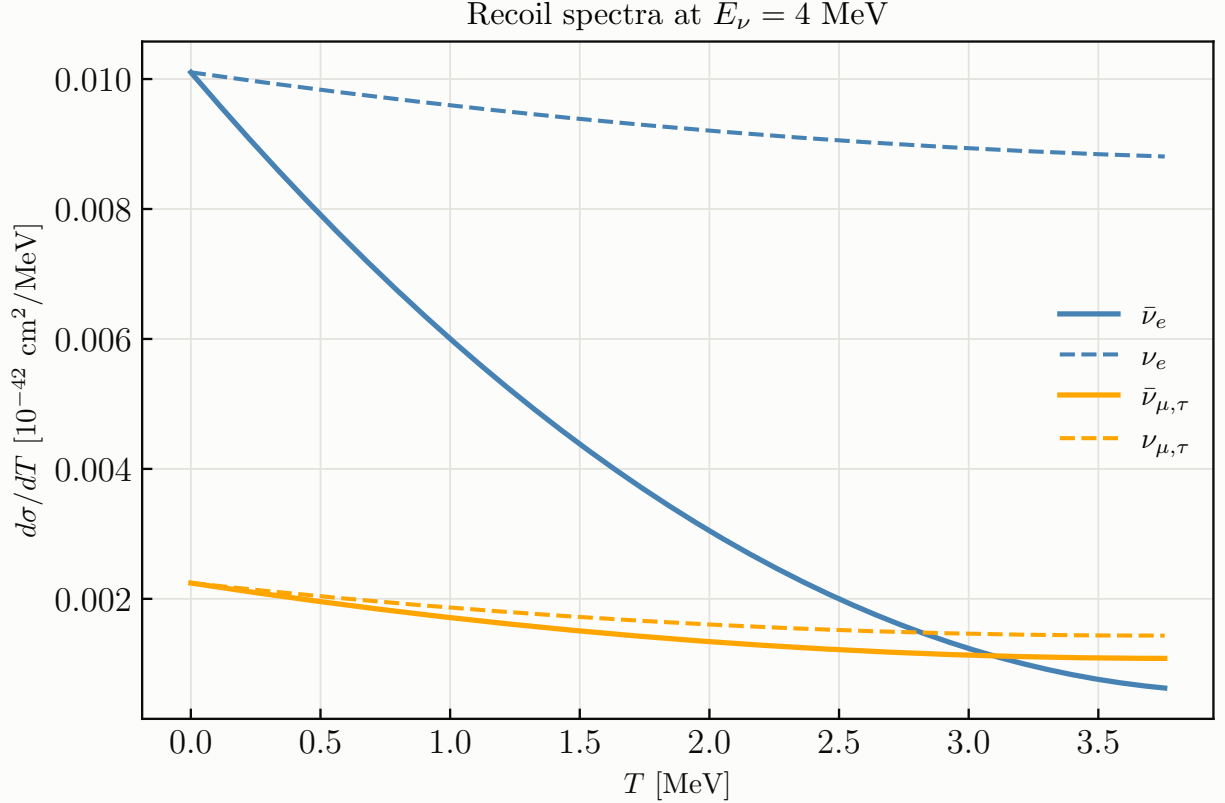


FIG. 2: Recoil spectra $d\sigma/dT$ at $E_\nu = 4$ MeV for the four flavour/helicity channels; the antineutrino–neutrino difference is the CC–NC interference.

${}^9\text{Li}/{}^8\text{He}$, world reactors, ${}^{214}\text{Bi-Po}$ and “other”, about 45 per day against 5.8×10^4 per day) enter at their release priors [7], and the shared ${}^{235}\text{U}$ shape modes act on both channels coherently, so the anchor constrains them in situ.

C. Results

Table I decomposes $\sigma(\sin^2 \theta_W)$ at 30 MW yr, and Fig. 4 shows the exposure dependence.

Two structural conclusions come out of this. First, *the $E\nu ES$ -only measurement is limited by the flux shape*: the recoil shape breaks the normalisation degeneracy far better than the naive rate argument ($|d \ln R / d \sin^2 \theta_W| \approx 4$) suggests, but the 2–4% correlated bin uncertainties of the measured ${}^{235}\text{U}$ spectrum imitate the coupling-induced shape change and saturate the precision at $\sigma(\sin^2 \theta_W) \approx 0.47\%$ regardless of exposure. Second, *the joint fit removes that wall*: with the normalisation freed and measured by the anchor, the ${}^{235}\text{U}$ shape covariance *and* the energy-scale pulls become exact nulls (the 7×10^7 -event anchor pins both), and what remains is set by the

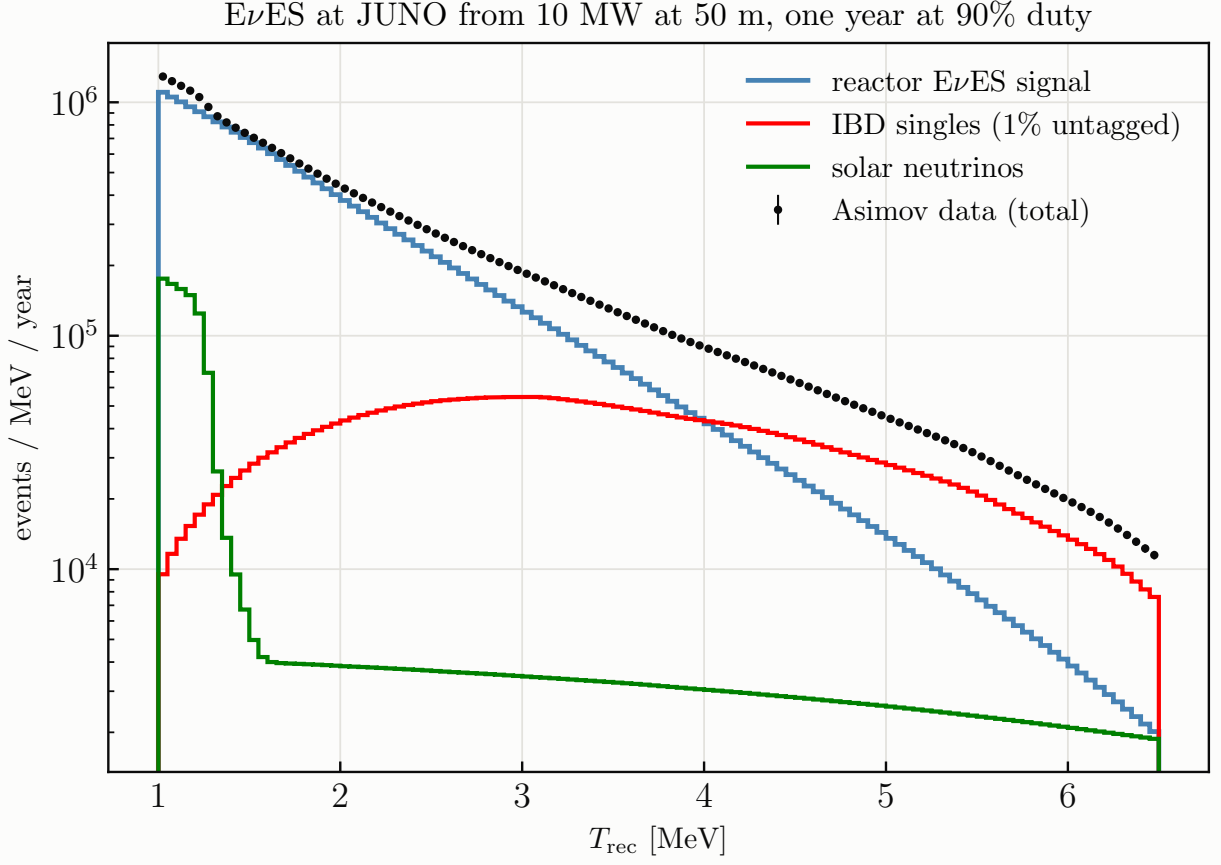


FIG. 3: Asimov $E\nu\text{ES}$ recoil spectrum with the modelled backgrounds, per calendar year at 90% duty, for 10 MW at 50 m ($\approx 1.0 \times 10^6$ signal, 1.9×10^5 IBD singles and 6×10^4 solar events per year in the window).

two transfer terms plus, at the +6% and +7% level, the fuel-evolution mode and the solar/JUNO backgrounds. The result, $\sigma(\sin^2 \theta_W) = 0.11\%$ (1.1×10^{-3}) at 30 MW yr and 0.054% at the plausible best of the transfer inputs, is a $\sim 0.25\%$ -of-itself low-energy weak mixing angle, competitive with the best existing determinations [10]. Figure 6 shows the same anchoring in the (g_V, g_A) plane: $\sigma(g_V) = 0.0027$ and $\sigma(g_A) = 0.0033$ at 30 MW yr.

Figure 5 places this in the existing landscape. Electron-flavour experiments see the CC amplitude too, so their cross sections depend on $(g_V + 1, g_A + 1)$ and constrain *bands* along the resulting degeneracies rather than ellipses: $\nu_e e$ (LSND [16], LAMPF [17]) bounds $(g_V + g_A + 2)^2 + (g_V - g_A)^2/3 = 2.34 \pm 0.35$, and $\bar{\nu}_e e$ (TEXONO [18]) bounds $(g_V - g_A)^2 + (g_V + g_A + 2)^2/3$ through $\xi = \sigma/\sigma_{\text{SM}} = 1.08 \pm 0.26$. Muon-flavour scattering (CHARM II [19]: $g_V = -0.035 \pm 0.017$, $g_A = -0.503 \pm 0.017$; BNL E734 [20]: -0.107 ± 0.045 , -0.514 ± 0.036) is pure NC and gives ellipses; the PDG world average, $g_V = -0.040 \pm 0.015$ and $g_A = -0.507 \pm 0.014$ [1], is domi-

$E\nu$ ES-only variant, 30 MW yr	$\sigma(\sin^2 \theta_W)(\%)$
all systematics	0.473
no flux-shape covariance	0.214
no solar/JUNO backgrounds	0.456
no energy scale/bias/res. pulls	0.468
no fuel-evolution mode	0.472
statistics only	0.013
joint IBD+ $E\nu$ ES, defaults (0.2% xsec, 0.5% ratio)	0.111
IBD xsec 0.5% (Strumia-Vissani)	0.136
channel ratio 0.2%	0.065
both at best (0.1%, 0.2%)	0.054
no ^{235}U shape covariance	0.106

TABLE I: Precision on $\sin^2 \theta_W$ in percentage for 30 MW yr near reactor exposure (≈ 3.3 calendar years with about 90% uptime).

nated by CHARM II and assumes flavour universality. Neutrino trident production (CCFR [21], $\sigma/\sigma_{\text{SM}} = 0.82 \pm 0.28$; CHARM II [22], 1.58 ± 0.57) constrains the ν_μ -*muon* couplings [23] rather than ν -*e*, so I do not draw it. Reactor $\bar{\nu}_e e$ data combined give $\sin^2 \theta_W = 0.252 \pm 0.030$ [24]. Our measurement is $\bar{\nu}_e$: it lands on TEXONO's band and shrinks it by two orders of magnitude, and its joint 30 MW yr ellipse (semi-axes 0.004×0.002) sits inside the CHARM II and world-average circles, 5–8 $\times$ tighter in each direction and from a different flavour. All contours are centred on the PDG world-average point.

III. STERILE-NEUTRINO WIGGLES ACROSS THE FINITE DETECTOR (NOTEBOOK 6)

A. The idea

For a point source at distance D from the centre of a fiducial sphere of radius R , vertices span $L \in [D - R, D + R]$ with the analytic density $dN/dL \propto 2\pi L^2 [1 - (L^2 + D^2 - R^2)/(2LD)]/L^2$. At $D = 50$ m the baseline spans 33–67 m, a factor of two. For $\Delta m_{41}^2 \sim 0.1\text{--}10\text{ eV}^2$ the oscillation length at reactor energies is metres to tens of metres, so the disappearance pattern *wiggles across the detector* (Figs. 7 and 8). Binning events in reconstructed vertex baseline L makes JUNO a many-baseline experiment in one fill, and no flux or cross-section systematic can follow: the

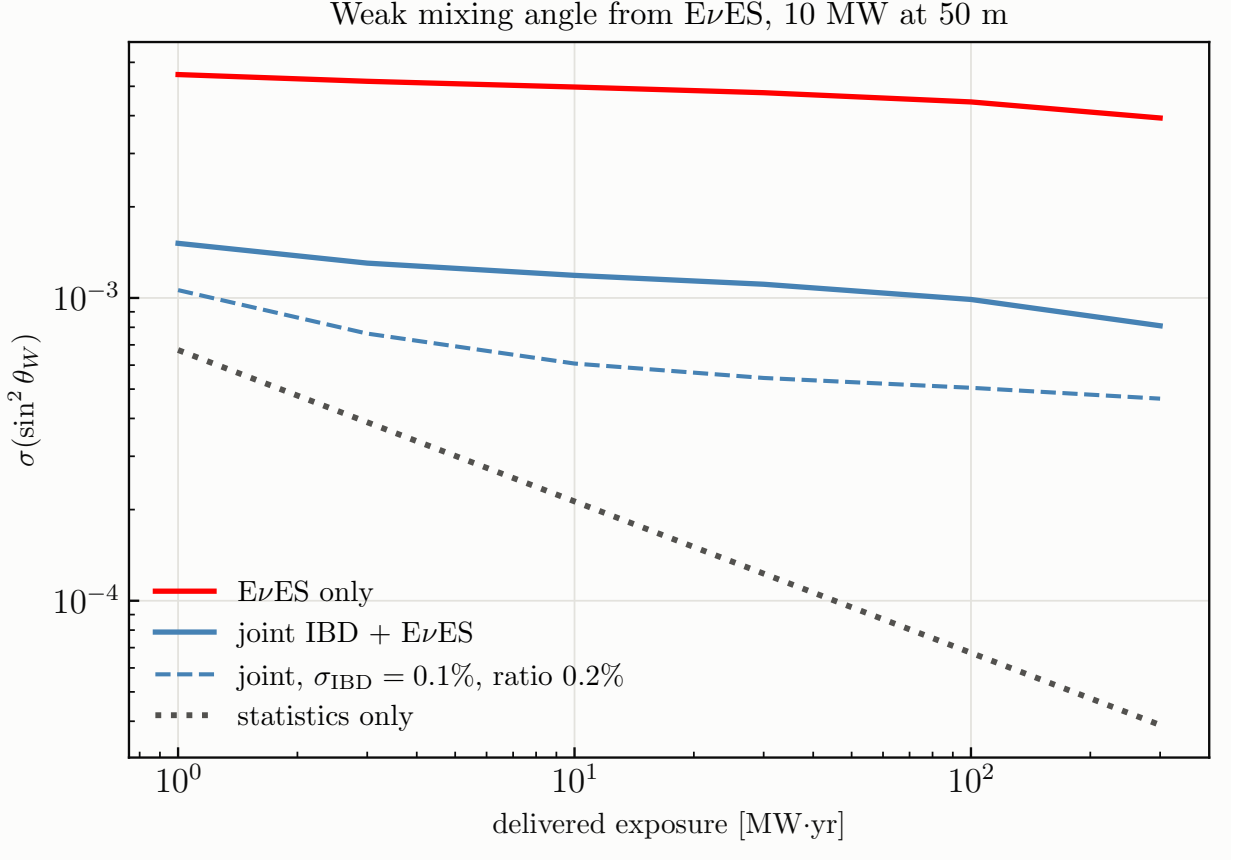


FIG. 4: $\sigma(\sin^2 \theta_W)$ versus delivered exposure. The $E\nu$ ES-only fit saturates on the measured flux-shape covariance; the IBD-anchored fit keeps converting statistics into precision.

geometry is exact and the source spectrum carries no L -dependence.

B. Oscillation, smearing, statistics

I use the full 3+1 vacuum survival probability,

$$P_{ee} = 1 - (1 - s_{14})^2 (1 - P_{ee}^{3\nu}) - 4s_{14}(1 - s_{14}) \sum_{i=1}^3 |U_{ei}|_{3\nu}^2 \sin^2 \Delta_{i4}, \quad s_{14} = \sin^2 \theta_{14}, \quad (6)$$

with the exact three-flavour $P_{ee}^{3\nu}$ in the Δm_{ee}^2 parameterisation [25] and $\Delta_{i4} = 1.267 (m_4^2 - m_i^2) L/E$ ($\Delta m_{31}^2 = \Delta m_{ee}^2 + s_{12}^2 \Delta m_{21}^2$). I validated it against a brute-force PMNS-amplitude computation to 8.5×10^{-6} . At 50 m it reduces to $1 - \sin^2 2\theta_{14} \sin^2 \Delta_{41}$, while at 1.4 km, where $\Delta_{31} \approx 1$ rad, the three sterile phases separate and the null hypothesis is the θ_{13} -oscillated spectrum. IBD (CC) and $E\nu$ ES (CC+NC) deplete identically, since an oscillated $\bar{\nu}_e$ is sterile.

Two effects smear the true baseline at fixed reconstructed vertex: the finite core (a uniform ball of radius a has line-of-sight variance $a^2/5$) and the vertex resolution. The oscillating term damps

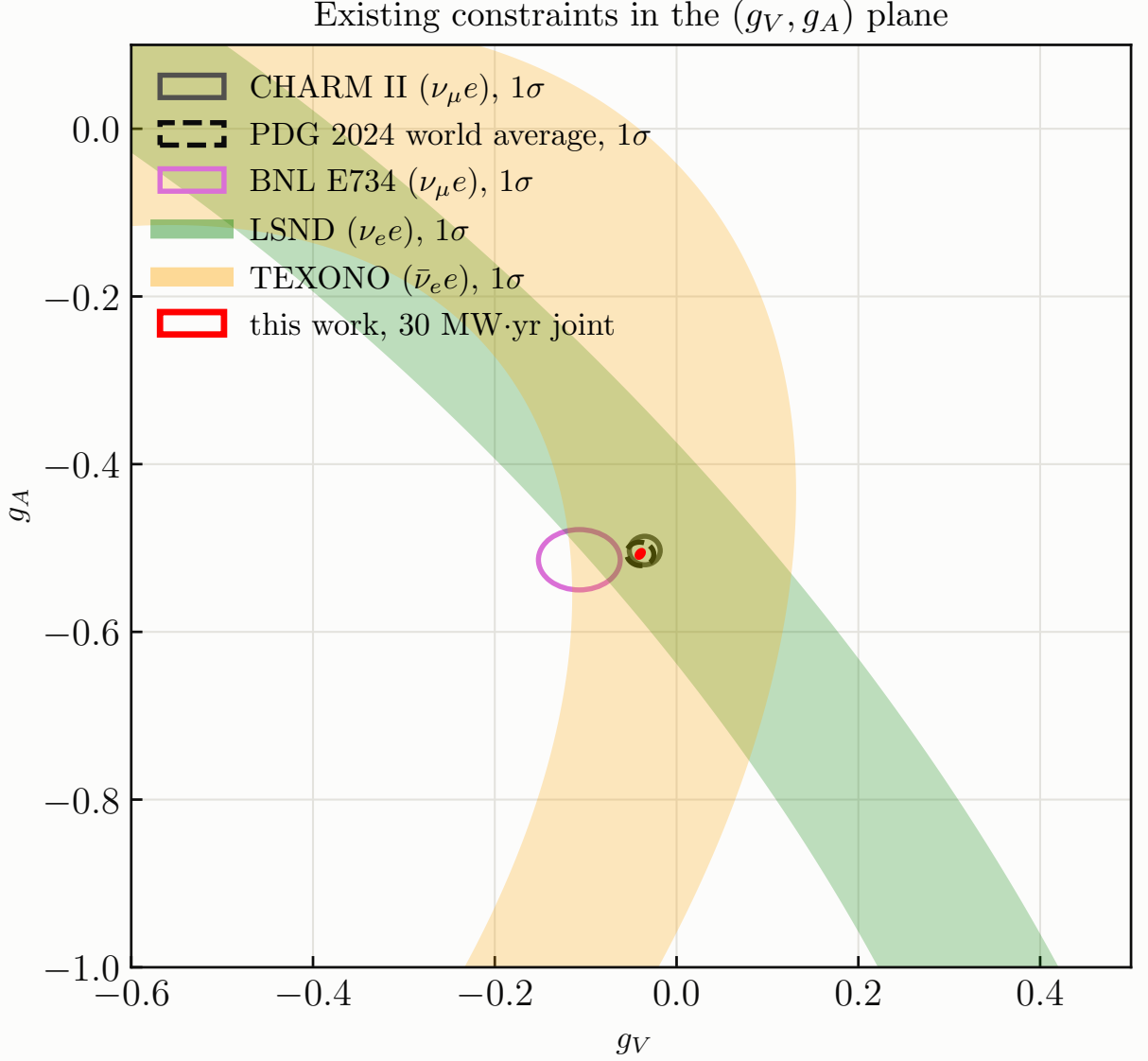


FIG. 5: Existing 1σ constraints in the (g_V, g_A) plane: the $\nu_e e$ (LSND) and $\bar{\nu}_e e$ (TEXONO) bands, the $\nu_\mu e$ ellipses (CHARM II, BNL E734) and the PDG world average, with this work's 30 MW yr joint region at the PDG point.

analytically, $\langle \sin^2 kL \rangle = \frac{1}{2}[1 - \cos(2kL)e^{-2k^2\sigma_L^2}]$ with $\sigma_L^2 = \sigma_1^2/E + a^2/5$ and $k = 1.267 \Delta m^2/E$; bin averaging is exact via sub-nodes. The prediction is linear in $\sin^2 2\theta_{14}$, so against the Asimov null $\Delta\chi^2 = \sin^4 2\theta_{14} Q(\Delta m^2)$ with $Q = W^T C^{-1} W$ for one wiggle template W per Δm^2 , and the entire exclusion curve is analytic (95% CL at $\Delta\chi^2 = 5.99$).

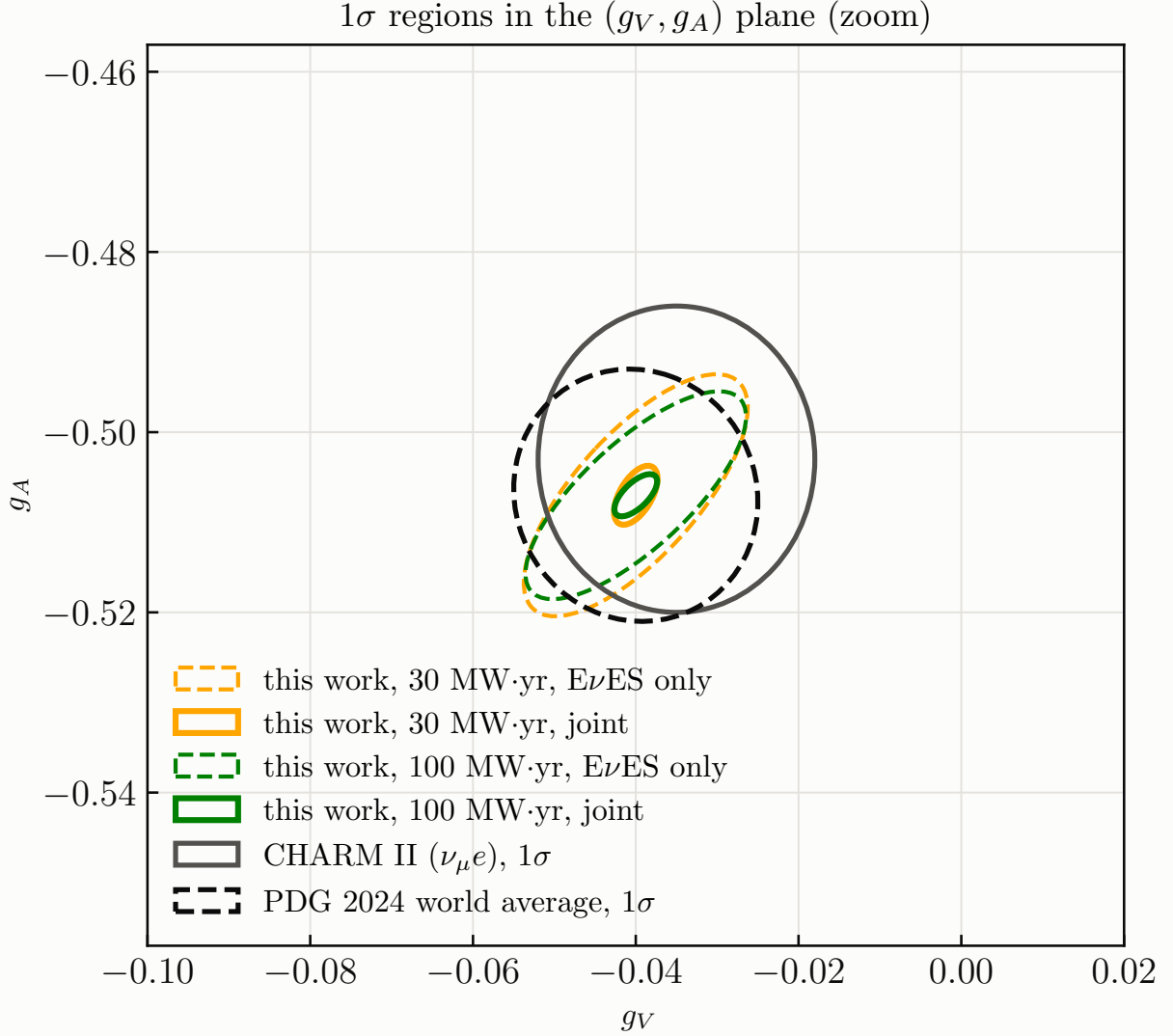


FIG. 6: Zoom on the PDG point: 1 σ Fisher regions of this work, E ν ES-only (dashed) and IBD-anchored (solid), against the CHARM II and PDG world-average ellipses.

C. Why the analysis is shape-only, and what that costs

The Daya Bay ^{235}U spectrum was *measured* at ~ 500 m: a sterile with $\Delta m_{41}^2 \sim 0.1\text{--}3\text{ eV}^2$ would have depleted it by $\approx \sin^2 2\theta_{14}/2$ in normalisation (4.8–5.0% at $\sin^2 2\theta_{14} = 0.1$, computed) and left a $\lesssim 1\%$ energy tilt. Using its normalisation or shape as a prior would inject the very signal being searched for. The standard analysis therefore leaves the flux normalisation *completely free* and the reactor E -shape *completely free* (one unconstrained mode per E and per T bin, coherent across L ; 263 modes in total). Only the L -direction wiggle structure at fixed E can then carry sterile information, and it carries essentially all of it: relative to a Daya-Bay-prior analysis the

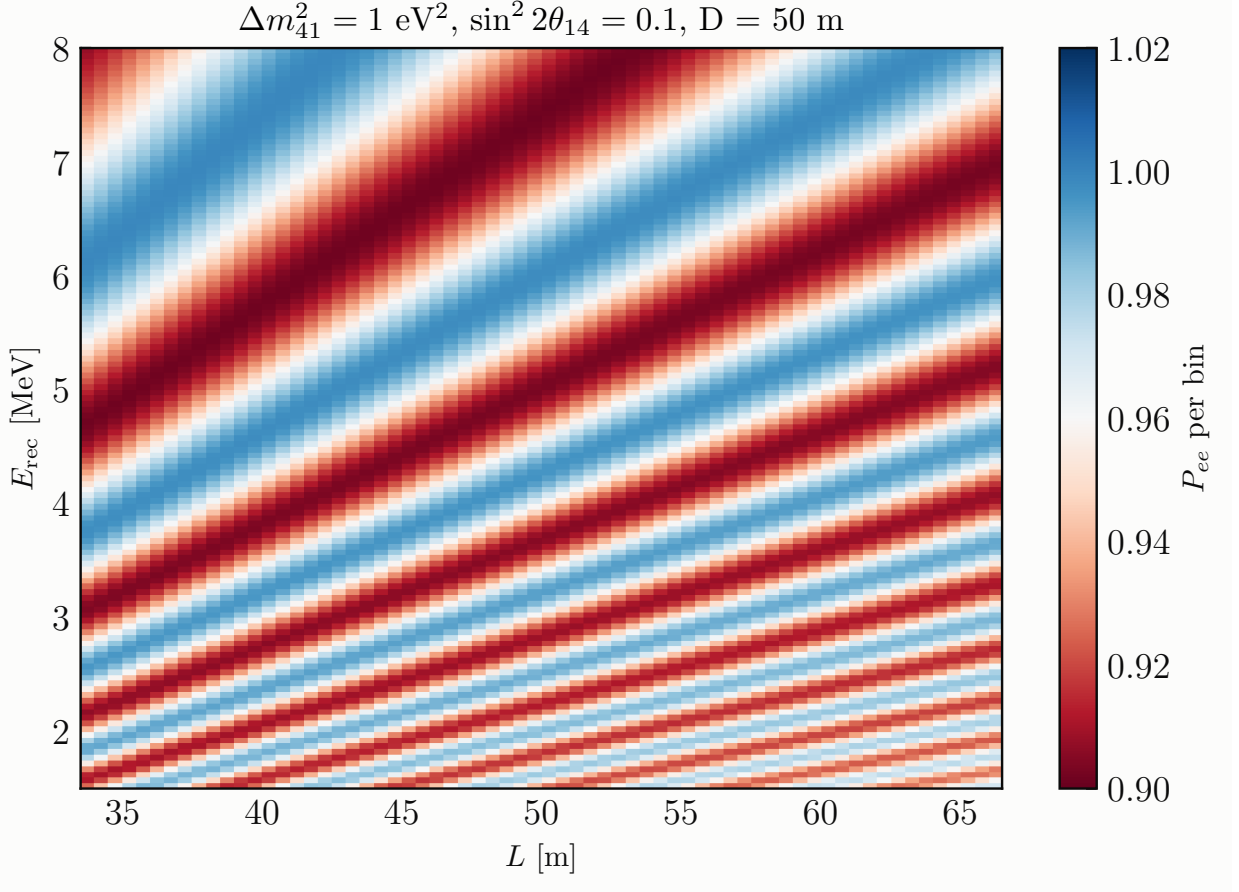


FIG. 7: Survival probability per (L, E_{rec}) bin for $\Delta m^2_{41} = 1 \text{ eV}^2$, $\sin^2 2\theta_{14} = 0.1$, $D = 50 \text{ m}$.

limits move by less than 0.5% across $0.03\text{--}3 \text{ eV}^2$, and by 11% only at the 10 eV^2 wall. The Daya Bay spectrum serves only as a shape template for event rates. The remaining nuisances are the detector response non-uniformity across the volume, modelled as a Gaussian-correlated field over L (0.5% amplitude, 1.5 m correlation length, the one systematic in the wiggle direction), the fuel evolution ($\pm 30\%$, a null), and the IBD/ $\text{E}\nu\text{ES}$ ratio (0.5%). The backgrounds of Section II (IBD singles, solar, and the standard JUNO components distributed as volume in L) are included and shift the limits by less than 0.1%: being smooth in both L and E , they cannot make wiggles.

D. Results

Figure 9 shows the 95% CL reach for 27 MW yr at each of five standoffs against the existing landscape (DANSS [26], PROSPECT [27], STEREO [28], RENO+NEOS [29], Gallium 2σ [30, 31], KATRIN 2025 [32], and the IsoDAR@Yemilab projection [33]). Figure 10 isolates what the finite size buys, and Fig. 11 the high- Δm^2 wall.

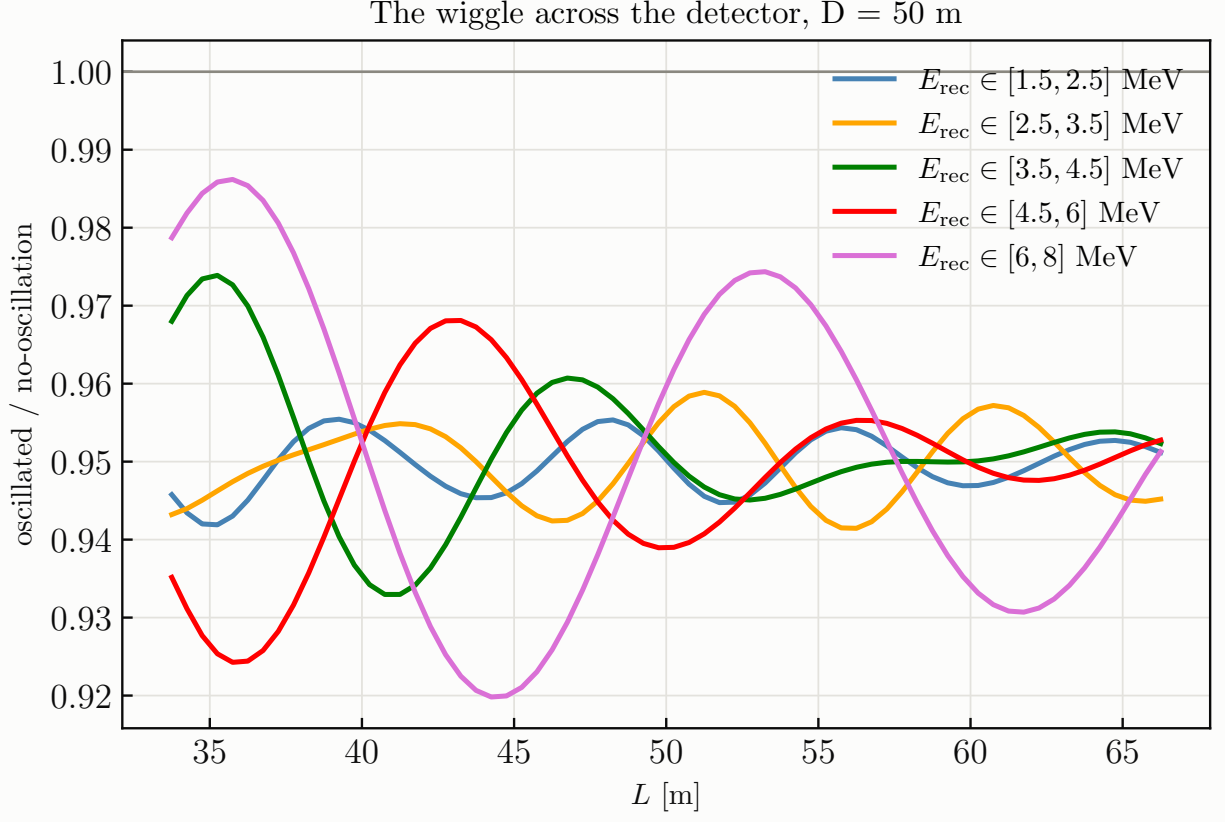


FIG. 8: The same benchmark: oscillated/unoscillated ratio versus L in five energy windows across the spectrum. The wavelength scales as E (~ 5 m at 2 MeV, ~ 20 m at 7 MeV), so different energies sample the same L range with different phase advance; this (L, E) pattern is what no source systematic can mimic.

	$D = 50$ m	100 m	500 m	1400 m
IBD rate [day^{-1}]	6.1×10^4	1.5×10^4	590	71
baseline span [m]	33–67	84–117	484–517	1384–1417
best $\sin^2 2\theta_{14}$ (95%)	$\approx 1 \times 10^{-3}$	$\approx 2 \times 10^{-3}$	$\approx 2 \times 10^{-2}$	$\approx 5 \times 10^{-2}$
at Δm_{41}^2 [eV^2]	0.3–1	0.1–0.5	0.05–0.1	0.01–0.03

TABLE II: Sterile reach per standoff, 27 MW yr each. Each standoff owns roughly a decade of Δm^2 (the reach follows the accumulated phase $\Delta_{41} \propto \Delta m^2 D/E$) while the rate falls as $1/D^2$.

The findings, in order of importance:

- The vertex-binned search reaches $\sin^2 2\theta_{14} \approx 10^{-3}$ over $0.2\text{--}3\text{ eV}^2$ at 50 m, an order of magnitude beyond the reactor short-baseline experiments and covering the entire Gallium

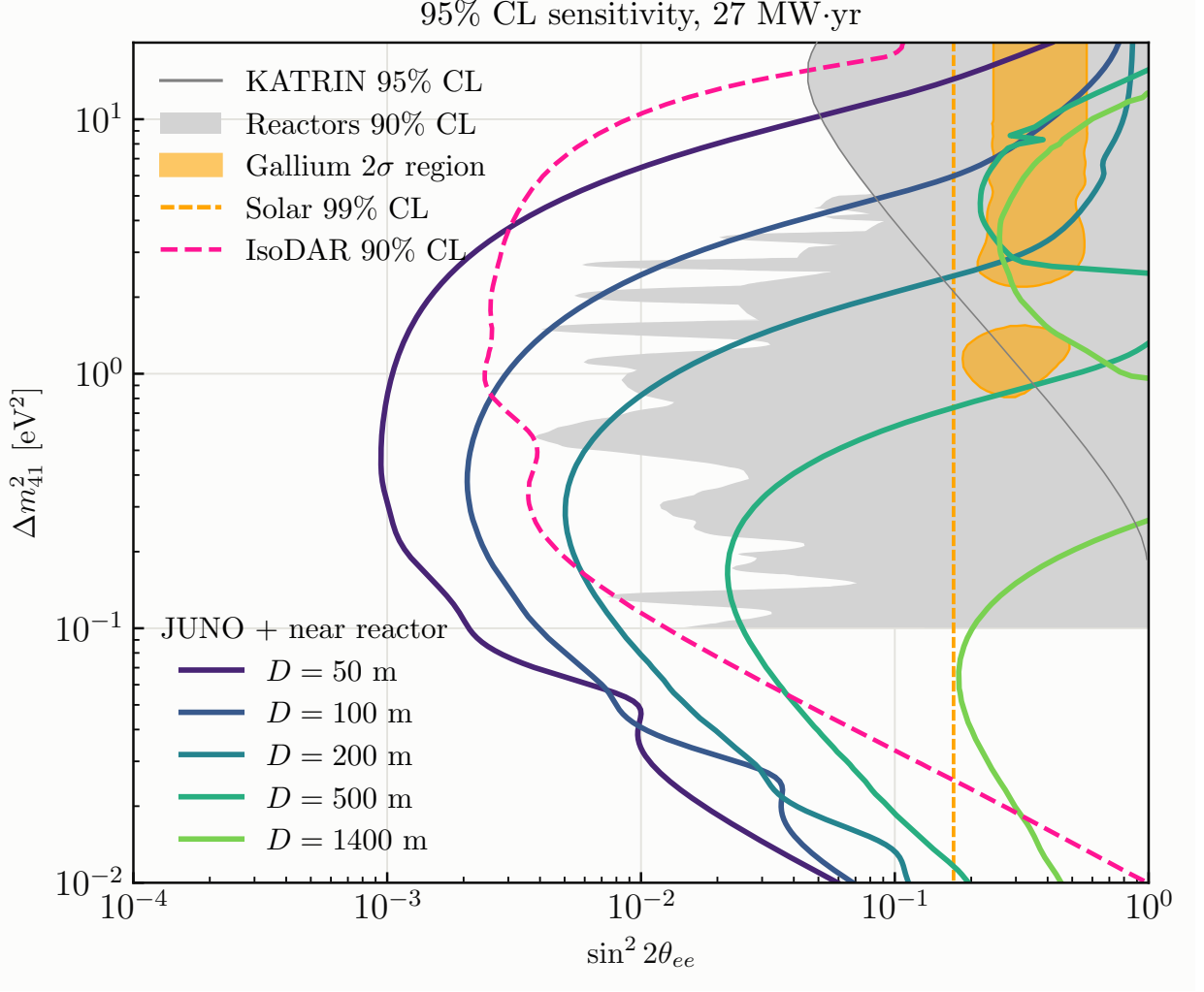


FIG. 9: 95% CL sensitivity in the $\bar{\nu}_e$ -disappearance plane for 27 MW yr at $D = 50, 100, 200, 500$ and 1400 m (each standoff separately), against the reactor short-baseline exclusions (DANSS, PROSPECT, STEREO, RENO+NEOS; grey), KATRIN 2025, the Gallium 2σ region, the solar bound, and the IsoDAR@Yemilab projection.

region. It beats the integrated analysis by a Δm^2 -dependent factor, about 5 at the integrated analysis's one sweet spot near 1 eV^2 (where its surviving E -wiggles resist the flux covariance) and up to ~ 70 elsewhere, and unlike it needs no flux prior at all. The integrated analysis's reach hinges on an accidental alignment between oscillation shape and flux uncertainty; the L -binned one is uniformly robust.

- Each standoff owns roughly a decade of Δm^2 (Table II). The compact standoffs are where the sensitivity is, and the kilometre ones open $\Delta m^2 < 0.1 \text{ eV}^2$ at the few-percent level. The full 3+1 formula matters at the few-percent level in the limit at 1.4 km for $\Delta m^2 \lesssim 0.1 \text{ eV}^2$

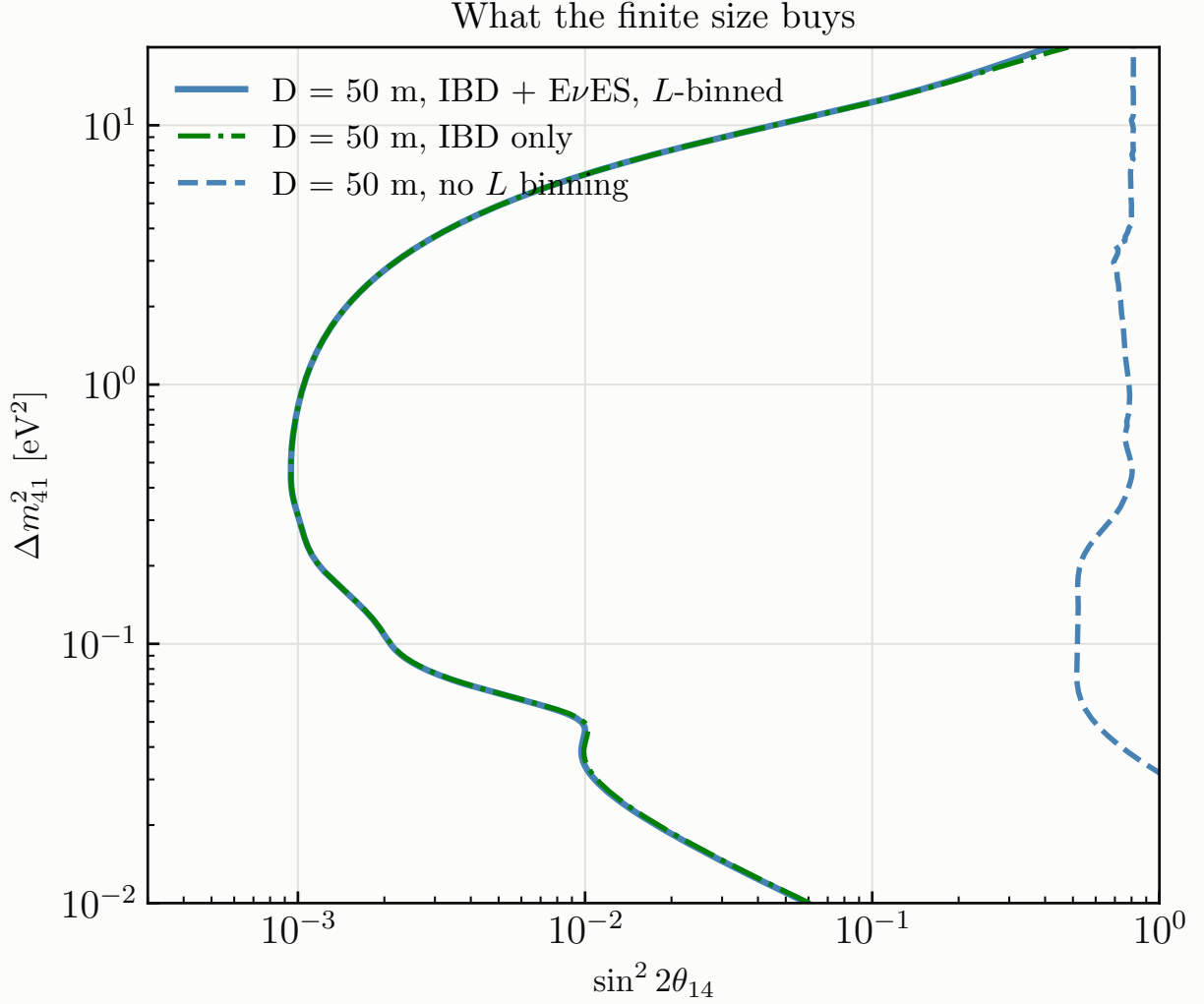


FIG. 10: The vertex-binned search against the same data analysed as a single integrated baseline, and the IBD-only variant.

(the Δ_{34} term dilutes the wiggle) and is identically null at 50 m.

- The high- Δm^2 wall is the *core size*: a 0.5 m core reaches $\sim 10 \text{ eV}^2$ and a 1.5 m core ~ 4 . JUNO's vertexing is never the bottleneck; even for a point source the *energy* resolution, through the L/E phase ($\sigma_\phi \propto \Delta m^2 L \sigma_E / E^2$), ends the search by 20–30 eV^2 , decaying in stages with the high- E tail surviving longest. Above the wall every configuration saturates at the rate-only floor $\sin^2 2\theta_{14} \approx 0.09$. Compactness is as valuable as power.
- The honest limiter in $\sin^2 2\theta_{14}$ is the response uniformity across the volume; the statistics floor lies a factor ~ 2 below the 0.5% assumption, and 0.1% would gain another factor of 2–3. $E\nu\text{ES}$ rides along but adds little: without E_ν reconstruction its wiggles wash out (the

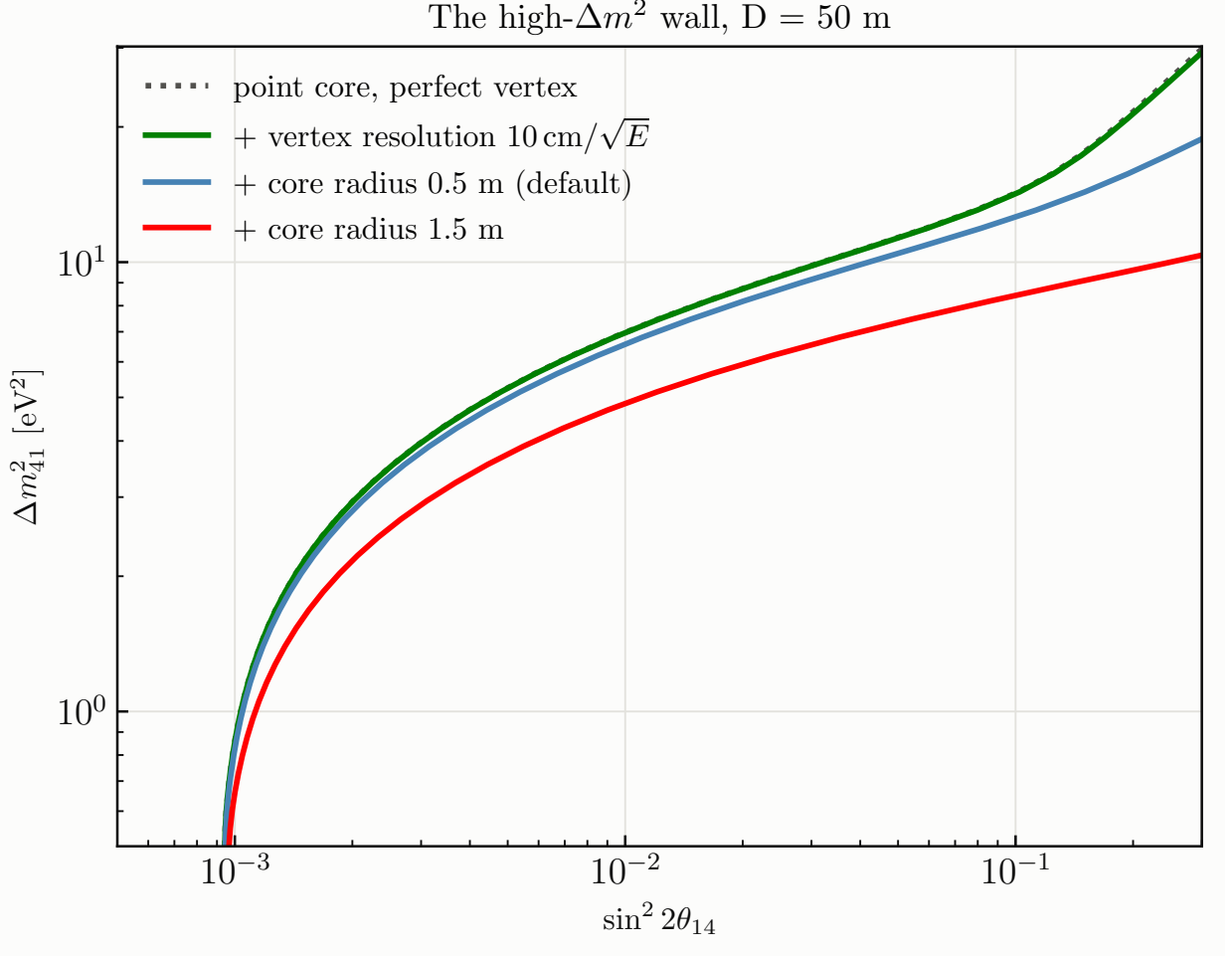


FIG. 11: The high- Δm^2 wall dissected: idealised point source and perfect vertexing, then the vertex resolution, then finite cores of 0.5 and 1.5 m.

true- T ratio varies by 0.027 peak-to-peak against 0.10 in the input).

IV. AXION-LIKE PARTICLES FROM THE CORE (NOTEBOOK 7)

A. Physics and formalism

The core is a photon source of $\sim 5 \times 10^{18} \gamma/\text{s}$: photons scattering in the fuel convert to ALPs by Primakoff ($g_{a\gamma\gamma}$, coherent Z^2 on uranium) or Compton-like (g_{aee}) processes, stream unimpeded through shielding, and reach JUNO, where they scatter back into visible photons or electrons through the inverse processes or decay in flight. I follow the calculation of Ref. [34] for liquid-scintillator detectors, with the underlying formulas of Refs. [35, 36]; every expression below is the one implemented in `reactor/alps.py`. The Lagrangian is $\mathcal{L} \supset -\frac{1}{4}g_{a\gamma\gamma} a F_{\mu\nu} \tilde{F}^{\mu\nu} - ig_{aee} a \bar{e}\gamma_5 e$, and

one coupling is switched on at a time.

a. Reactor photon flux. The prompt continuum is the FRJ-1 power law [35, 37]

$$\frac{d\Phi_\gamma}{dE_\gamma} = 5.8 \times 10^{17} \left(\frac{P}{\text{MW}} \right) e^{-1.1 E_\gamma / \text{MeV}} \text{MeV}^{-1} \text{s}^{-1}, \quad (7)$$

which integrates to $1.75 \times 10^{18} \gamma/\text{s}$ above 1 MeV at 10 MW (the analytic $5.8 \times 10^{18} e^{-1.1}/1.1$). MINER's simulated flux is $\sim 20\times$ larger, so Eq. (7) is the conservative choice, and the one made by the reference analyses.

b. Primakoff production and inverse-Primakoff detection. For $\gamma(p_1) + N(p_2) \rightarrow a(k_1) + N(k_2)$ on a nucleus of mass M , charge Z , the exact $2 \rightarrow 2$ cross section [35, 38] is

$$\frac{d\sigma_P}{dt} = 2\alpha Z^2 F^2(t) g_{a\gamma\gamma}^2 \frac{M^4}{t^2 (M^2 - s)^2 (t - 4M^2)^2} \left\{ m_a^2 t (M^2 + s) - m_a^4 M^2 - t [(M^2 - s)^2 + st] \right\}, \quad (8)$$

integrated over $t \in [t_{\min}, t_{\max}]$, $t_{\max/\min} = m_a^4/4s - (p_{1\text{cm}} \mp k_{1\text{cm}})^2$ with $p_{1\text{cm}} = (s - M^2)/2\sqrt{s}$, $k_{1\text{cm}} = [(s + m_a^2 - M^2)^2/4s - m_a^2]^{1/2}$, $s = M^2 + 2ME_\gamma$, on a logarithmic t -grid (the integrand is sharply forward-peaked). The atomic screening form factor is the Thomas–Fermi–Molière form $F^2(t) = [a_0^2 |t| / (1 + a_0^2 |t|)]^2$, $a_0 = 184.15 e^{-1/2} / (Z^{1/3} m_e)$, essential below $m_a \sim 10$ keV; the nuclear form factor is negligible for $E_a \lesssim 10$ MeV. Because $|t|/M^2 \ll 1$ the ALP carries the photon energy, $E_a \simeq E_\gamma$, so the emitted flux is

$$\frac{d\Phi_a}{dE_a} = \frac{\sigma_P(E_a)}{\sigma_{\text{SM}}(E_a)} \frac{d\Phi_\gamma}{dE_\gamma} \Big|_{E_\gamma=E_a}, \quad (9)$$

where σ_{SM} (Rayleigh + Compton + photoelectric + pair) is the total photon cross section in uranium from XCOM [39]; the branching ratio $\sigma_P/(\sigma_P + \sigma_{\text{SM}})$ is $\sigma_P/\sigma_{\text{SM}}$ to excellent accuracy in the coupling range of interest. Detection by inverse Primakoff on carbon and hydrogen uses $\sigma_P^{\text{det}} = 2\sigma_P$ (spin counting: a spin-0 initial state instead of spin-1).

c. Compton-like production and inverse-Compton detection. For $\gamma e \rightarrow ae$ with $s = m_e^2 + 2E_\gamma m_e$, threshold $s > (m_a + m_e)^2$ i.e. $E_\gamma^{\text{th}} = m_a + m_a^2/2m_e$, and $x = 1 - E_a/E_\gamma + m_a^2/2E_\gamma m_e$ [35, 40]:

$$\frac{d\sigma_C}{dx} = \frac{\alpha g_{aee}^2 x}{4(s - m_e^2)(1 - x)} \left[x - \frac{2m_a^2 s}{(s - m_e^2)^2} + \frac{2m_a^2}{(s - m_e^2)^2} \left(\frac{m_e^2}{1 - x} + \frac{m_a^2}{x} \right) \right], \quad (10)$$

integrated over $x_{\min, \max} = [(s - m_e^2)(s - m_e^2 + m_a^2) \mp (s - m_e^2) \sqrt{(s - m_e^2 + m_a^2)^2 - 4sm_a^2}] / 2s(s - m_e^2)$; the ALP flux is the convolution of $\sigma_C^{-1} d\sigma_C/dE_a \times \sigma_C/\sigma_{\text{SM}}$ with the photon flux (Ref. [34], Eq. 14). Detection by inverse Compton $ae \rightarrow \gamma e$ uses (Ref. [34], Eq. 22)

$$\frac{d\sigma_{IC}}{dE_\gamma} = \frac{\alpha g_{aee}^2}{32\pi} \frac{y}{m_e^2 p_a^2 E_\gamma} \left(1 + \frac{4m_e^2 E_\gamma^2}{y^2} - \frac{4m_e E_\gamma}{y} - \frac{4m_a^2 m_e p_a^2 E_\gamma \sin^2 \theta}{y^3} \right), \quad y = 2m_e E_a + m_a^2, \quad (11)$$

integrated over $E_\gamma \in [y/2(m_e + E_a + p_a), y/2(m_e + E_a - p_a)]$ with $\cos \theta = (m_e + E_a - y/2E_\gamma)/p_a$. Axio-electric absorption is neglected: it scales as Z^5 and on carbon at MeV energies is $\sim 10^{-4}$ of the germanium rates of the reference analyses, negligible against inverse Compton here.

d. Decays, survival and decay-in-detector probabilities.

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}, \quad \Gamma(a \rightarrow e^+e^-) = \frac{g_{aee}^2 m_a}{8\pi} \sqrt{1 - \frac{4m_e^2}{m_a^2}} \quad (m_a > 2m_e), \quad (12)$$

with lab decay length $\ell_a = (p_a/m_a) \hbar c / \Gamma_a$. An ALP produced in the core survives to the detector centre with $P_{\text{surv}} = e^{-L/\ell_a}$ (for scattering, evaluated at $L = D = 50$ m) or to the front face with $e^{-(D-R)/\ell_a}$, and then decays inside with probability

$$P_{\text{decay}} = 1 - \exp(-\bar{\ell}/\ell_a), \quad \bar{\ell} = \frac{4}{3}R = 22 \text{ m}, \quad (13)$$

the mean chord of the fiducial sphere (this replaces the fixed depth $r = 0.7$ m of the JUNO-TAO geometry). The decay-length boundary of Ref. [34], Eq. (53), corrected to $g^2 = 8\pi\hbar c/(Lm_a\beta)$, is reproduced numerically to 10^{-4} .

e. Event yields. Over a live time Δt (27 MW yr = 3 yr at 90% duty), with $\mathcal{A} = \pi R^2 = 855 \text{ m}^2$ the detector face and $N_{C,H,e}$ the fiducial target counts,

$$N_{\text{scat}}^P = \frac{\Delta t}{4\pi D^2} \int dE_a \frac{d\Phi_a}{dE_a} e^{-D/\ell_a} 2[N_C \sigma_P^C(E_a) + N_H \sigma_P^H(E_a)], \quad (14)$$

$$N_{\text{scat}}^{IC} = \frac{\Delta t}{4\pi D^2} \int dE_a \frac{d\Phi_a}{dE_a} e^{-D/\ell_a} N_e \sigma_{IC}(E_a), \quad (15)$$

$$N_{\text{decay}} = \frac{\mathcal{A} \Delta t}{4\pi D^2} \int dE_a \frac{d\Phi_a}{dE_a} e^{-(D-R)/\ell_a} [1 - e^{-\bar{\ell}/\ell_a}], \quad (16)$$

with E_a integrated over the single-hit window 3–10 MeV. Compared with JUNO-TAO we trade a GW reactor for 10 MW and 30 m for 50 m, but gain $\sim 2 \times 10^4 \times$ the target mass and $\sim 25 \times$ the decay path; the decay term (16) is what carries the reach.

f. Validation. I checked every ingredient. $\Gamma(a \rightarrow \gamma\gamma)$ and $\Gamma(a \rightarrow e^+e^-)$ agree with analytic evaluation to 10^{-4} and vanish below the e^+e^- threshold. The exact Primakoff form (8) agrees with an independent implementation of the Dent et al. forward/recoilless form ($d\sigma/d\cos\theta = \frac{1}{4}g^2\alpha Z^2 F^2 |\vec{p}_a|^4 \sin^2\theta/t^2$) at the 6–40% level, with the largest deviations near threshold and at high energy, which is exactly the deviation Ref. [35] (footnote 5) reports for that approximation. The Compton cross section has the right threshold and positivity, and the right magnitude ($\alpha g^2/m_e^2$, i.e. of order Klein–Nishina at $g_{aee} = 1$: 3.3×10^{-25} against $2.1 \times 10^{-25} \text{ cm}^2$ at 1 MeV). The uranium σ_{SM} reproduces its XCOM anchors, and the flux integral its analytic value.

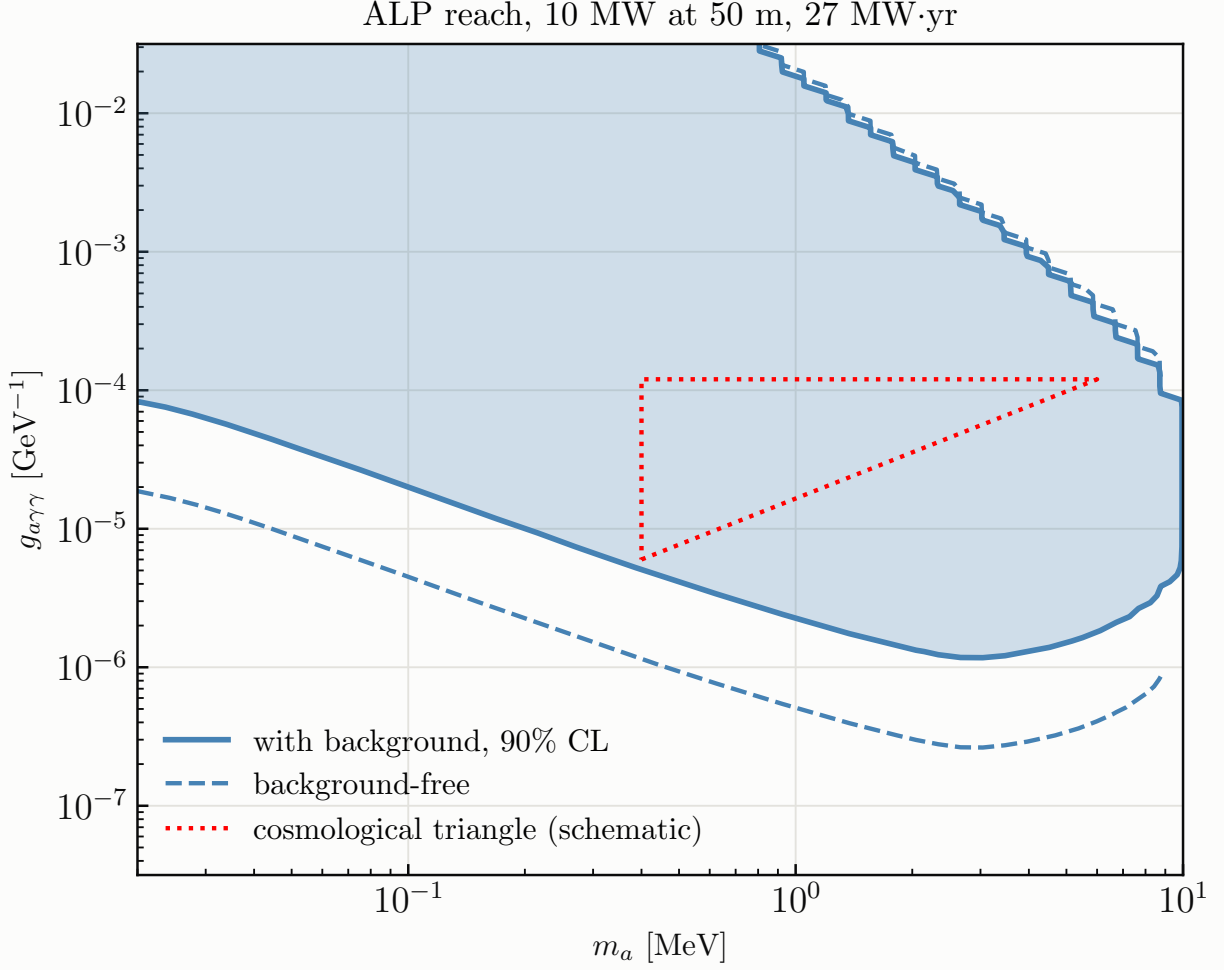


FIG. 12: 90% CL exclusion in the ALP–photon plane for 27 MW yr, with the full background (solid, filled) and background-free (dashed). The dotted wedge is the schematic cosmological triangle of Ref. [35], Fig. 10.

g. Background and statistics. The single-hit window is 3–10 MeV. The background is our own model of Section II: reactor $\text{E}\nu\text{ES}$ 3.3×10^5 , IBD singles 3.6×10^5 and solar $\text{E}\nu\text{ES}$ 3.1×10^4 events over the exposure (7.3×10^5 in total). Cosmogenics (^{12}B and friends) are not modelled; since $\chi^2 \propto g^8/N_b$, the reach in the coupling scales only as $N_b^{1/8}$, and even a $4\times$ larger background costs 20%. Following Ref. [34], Eq. (54), the single-bin statistic is $\chi^2 = N_s^2/(N_s + N_b + \sigma_{\text{sys}}^2 N_s^2)$ with $\sigma_{\text{sys}} = 0.1$, and 90% CL is $\Delta\chi^2 = 4.61$ (2 dof). The background-free limit is shown alongside, since the prompt/delayed photon and vertex handles of a LS detector could approach it.

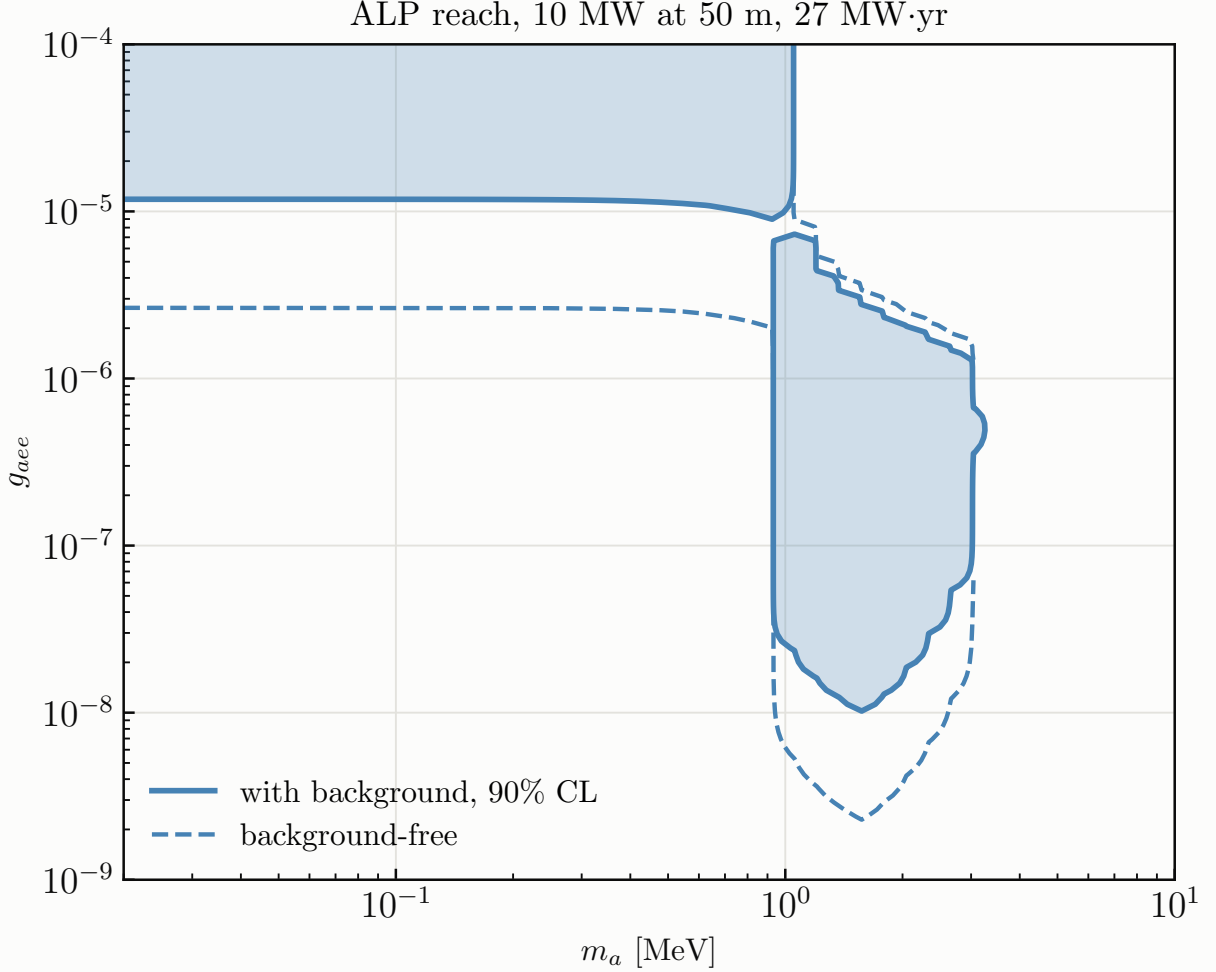


FIG. 13: Same in the ALP–electron plane; above the e^+e^- threshold the decay channel opens the band down to $g_{aee} \sim 10^{-8}$.

B. Results

The decay channel dominates by orders of magnitude at MeV masses (Fig. 12): at $m_a = 1$ MeV and $g_{a\gamma\gamma} = 10^{-5} \text{ GeV}^{-1}$ there are 7×10^5 decays against 0.06 scatterings over the exposure. With the background included, at $m_a = 1$ MeV the excluded band is $g_{a\gamma\gamma} \in [2.3 \times 10^{-6}, 1.4 \times 10^{-2}] \text{ GeV}^{-1}$, so the schematic cosmological triangle, the wedge at $m_a \sim 0.3\text{--}5$ MeV and $g_{a\gamma\gamma} \sim 10^{-5}\text{--}10^{-4} \text{ GeV}^{-1}$ between the beam-dump, HB-star and SN1987A exclusions where only cosmology-dependent bounds apply [41, 42], is covered in full. For scale, JUNO-TAO with background reaches $g \sim \text{few} \times 10^{-4}$ [34]; the near-reactor plus JUNO combination goes an order of magnitude deeper at the same reactor-off cost. In the electron plane (Fig. 13), at $m_a = 1.5$ MeV the excluded band is $g_{aee} \in [1.2 \times 10^{-8}, 3.1 \times 10^{-6}]$ with background, so the unexplored stripe that

Ref. [34] identifies is reached without the background-free assumption TAO needs. The caveats are the FRJ-1 flux, the absence of cosmogenics, single-bin counting (an energy-shape fit would only help), the pure-uranium core, the neglect of ALP attenuation in the shielding, and the schematic triangle overlay.

V. DISCUSSION AND CAVEATS

The three studies share a lesson about what a parked reactor’s *size and proximity* do for a 20 kt detector. For the SM couplings the win is the IBD anchor: $20\times$ the $\text{E}\nu\text{ES}$ statistics with the *same* flux, cancelling normalisation, flux shape and energy response against the 0.2% IBD cross section. For sterile oscillations the win is the detector’s finite size, which converts the oscillation into an L -direction observable that no source systematic can mimic, and which crucially allows a shape-only analysis independent of the Daya Bay measurement that would otherwise carry the signal. For ALPs the win is the decay volume: an 855 m^2 face and a 22 m chord watching a $5 \times 10^{18}\text{ }\gamma/\text{s}$ source.

The stated caveats are real. Radioactivity and cosmogenic singles are unmodelled in the $\text{E}\nu\text{ES}$ and ALP windows (they are severe below $\sim 3\text{ MeV}$ and would enter the ALP reach as $N_b^{1/8}$). The reactor spectrum is undefined below the IBD threshold. The response uniformity (0.5% over the volume) is the assumed limiter of the sterile search and should be replaced by JUNO’s calibration numbers. A companion feasibility estimate (notebook 8) finds that the reactor’s own neutron and gamma leakage require a research-reactor-class shield ($\sim 6\text{ m}$ water-equivalent, or $\sim 2\text{ m}$ steel plus a sub-metre high- Z collar, on top of JUNO’s 4 m of pool water) for the near-reactor programme not to spoil single-hit physics; this is exactly what a TRIGA-style pool installation provides. Finally, the mobile-reactor θ_{13} study (notebook 2) remains the strongest oscillation argument for such a source: the fixed-reactor variant (notebook 2b) shows that the 33 m detector provides only 0.026 rad of θ_{13} phase and cannot substitute for moving the source. The same finite size that is the right tool at metre oscillation lengths is the wrong one at kilometres.

Appendix A: Daya Bay flux model and covariance

a. The measured spectra. Daya Bay’s unfolding [3] releases the per-fission IBD yields Φ_i^0 [$\text{cm}^2\text{ fission}^{-1}$] in $n = 25$ neutrino-energy bins of width 0.25 MeV over $E_\nu \in [1.8, 8.0]\text{ MeV}$, for three spectra (^{235}U , ^{239}Pu , and the “Total”, their cycle-averaged mixture at fission fractions 0.564 :

0.076 : 0.304 : 0.056), with a 75×75 covariance V over the stacked $[^{235}\text{U}, ^{239}\text{Pu}, \text{Total}]$ vector. The near-reactor studies of this report use the ^{235}U and ^{239}Pu yields directly with the 50×50 block; the standard JUNO analysis (below) uses the Total.

Figure 14 shows the measured ^{235}U and ^{239}Pu yields with their released 1σ uncertainties (the diagonal of V ; the last bin spans 7.8–9.5 MeV), converted into IBD rates at JUNO for a hypothetical pure 10 MW core of each isotope at 50 m, together with the Huber–Mueller ^{238}U and ^{241}Pu spectra used for the minority components. The Huber–Mueller polynomial fits are valid only up to 8.5 MeV; above it the code applies an artificial exponential cutoff (marked in the figure), which is why the ^{238}U and ^{241}Pu curves kink there. This is immaterial, since these are 1–2% components and contribute nothing to any analysis above 8 MeV. Figure 15 is the companion: the actual HALEU core at the start ($\beta = 0$), middle and end ($\beta = 1$) of its cycle, built from those ingredients through Eqs. (1) and (2), with a 5% rate drop and a slight softening ($2.00 \rightarrow 1.90 \times 10^7$ IBD per year); its bottom panel shows the shape change as a ratio to the start of cycle. Figure 16 is the same figure for a commercial-LEU core, the Daya Bay measured fission fractions across their cycle ($F_{235} = 0.66 \rightarrow 0.49$, $F_{239} = 0.23 \rightarrow 0.36$), at the same power and distance. The LEU core has a 5% lower absolute rate (plutonium-rich fuel yields fewer $\bar{\nu}_e$ per unit power), and because the HALEU model applies Daya Bay’s Pu *increments* on top of a ^{235}U -dominated fresh core, its end-of-cycle distortion ($\sim 10\%$ at 8 MeV) is essentially the LEU one. That is the statement “follows the Daya Bay trajectory” made visible, and the reason it is a conservative model for a 2%- ^{238}U core, whose real plutonium breeding is slower.

b. From binned yields to a continuous flux (the NuFit prescription). Following Appendix A of the NuFit analysis that the repository’s standard method reproduces, the continuous flux times cross section is written as a smooth reference shape times a modulation,

$$S(E) = \phi_{\text{HM}}(E) \sigma_{\text{IBD}}(E) \sum_{m=1}^n y_m \delta_m(E), \quad (\text{A1})$$

where ϕ_{HM} is the Huber–Mueller spectrum at the Daya Bay average fission fractions (with no isotope correction), σ_{IBD} is Vogel–Beacom, and $\{\delta_m\}$ are the cardinal cubic splines through the 25 bin centres ($\delta_m(E_{m'}) = \delta_{mm'}$). The coefficients are fixed by demanding that the bin-averaged prediction reproduce the measured yields exactly:

$$M_{im} = \frac{1}{\Delta E_i} \int_{\text{bin } i} \phi_{\text{HM}} \sigma_{\text{IBD}} \delta_m dE, \quad \mathbf{y}^0 = M^{-1} \Phi^0, \quad (\text{A2})$$

with the integrals done on a dedicated 1 keV grid; the residual $\max_i |(M\mathbf{y}^0)_i - \Phi_i^0|/\Phi_i^0$ is 6×10^{-16} . Above the last measured bin (9.5 MeV after the release’s extension) the modulation is held constant.

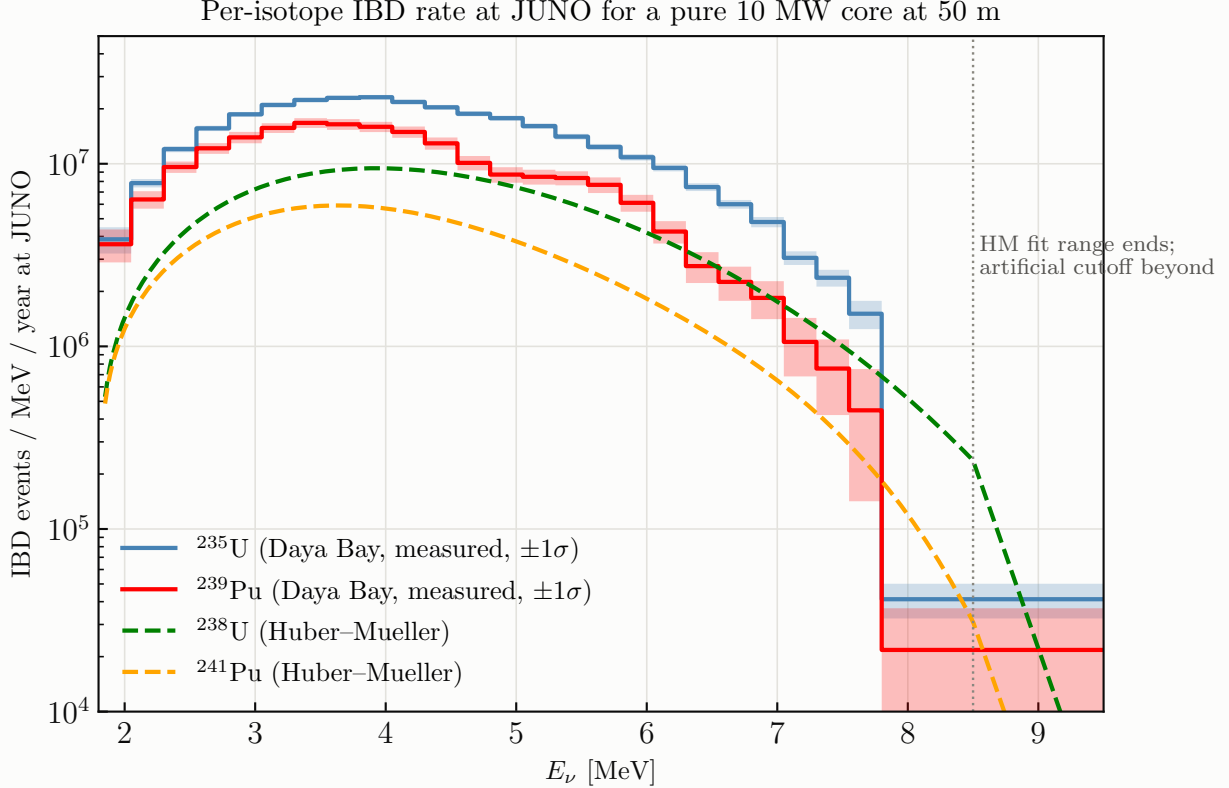


FIG. 14: Per-isotope IBD rate at JUNO for a pure 10 MW core of each isotope at 50 m: the measured Daya Bay ^{235}U and ^{239}Pu yields [3] with their released 1σ uncertainties as bands, and the Huber–Mueller [5, 6] ^{238}U and ^{241}Pu spectra (dashed). The dotted line marks the end of the Huber–Mueller fit range.

c. From covariance to pulls. The covariance is recast as n unit-Gaussian pulls by eigendecomposition, $V = O D O^T$, $\Psi = O\sqrt{D}$ (columns are modes), and each mode is pushed through the same interpolation, $\mathbf{y}^{(k)} = M^{-1}\Psi_{\cdot k}$, so that a pull ξ_k distorts the flux by

$$S(E) \rightarrow S(E) \left[1 + \sum_k \xi_k \frac{\sum_n y_n^{(k)} \delta_n(E)}{\sum_n y_n^0 \delta_n(E)} \right], \quad \chi^2 \supset \sum_k \xi_k^2, \quad (\text{A3})$$

which is exact to first order and keeps the pull penalty diagonal. In the near-reactor modules the same construction is applied to the relative covariance $V_{ij}/(\Phi_i^0 \Phi_j^0)$ of the ^{235}U block (and, in Section II, jointly to the 75×75 matrix so that a mode distorts the ^{235}U , ^{239}Pu and Total spectra *coherently*, since the far and near fluxes are correlated because they are anchored to the same measurement); the modes are then interpolated linearly in E and enter the Asimov covariance $C = \text{diag}(\mu) + \sum_k v_k v_k^T$ as $v_k = t \partial \mu / \partial \xi_k$. Modes with eigenvalue below 10^{-10} of the largest are dropped. In the sterile analysis of Section III this prior is deliberately *not* used, and the flux shape

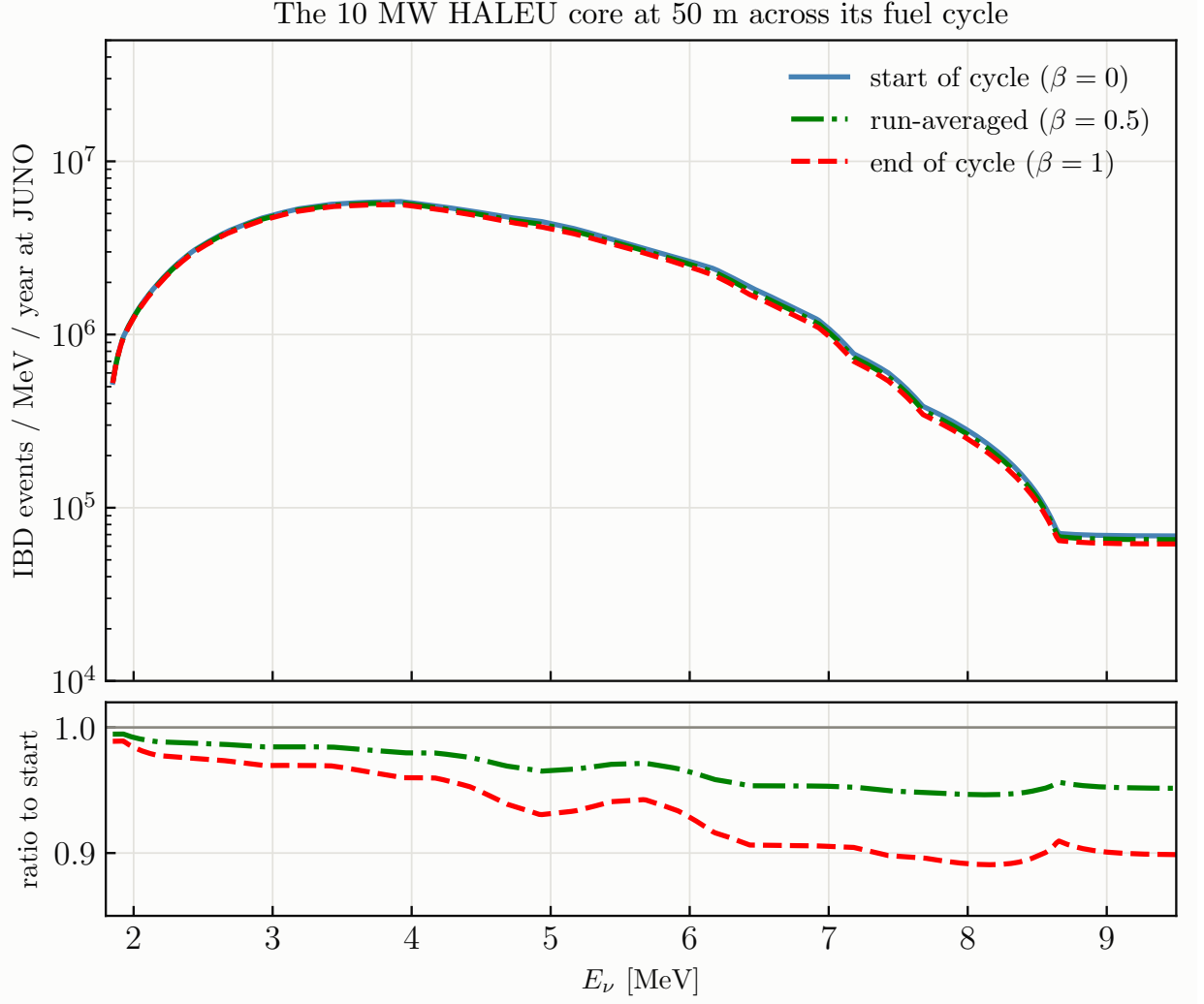


FIG. 15: The 10 MW HALEU core at 50 m at start, middle and end of its fuel cycle (Eq. (2)), built from the spectra of Fig. 14; bottom: ratio to the start of cycle.

and normalisation are left free, for the reason explained there.

d. The measured Total flux and JUNO's own spectrum. For the standard far-reactor analysis, the NuFit prescription additionally rescales the prediction bin-per-bin to JUNO's recovered un-oscillated spectrum,

$$k_i = \frac{(N_i^{\text{obs}} - B_i)/\bar{P}_{ee,i}}{N_i^{\text{pred,unosc}}}, \quad (\text{A4})$$

where $\bar{P}_{ee,i}$ is JUNO's own released model survival probability per bin; this pins the un-oscillated shape to data and turns flux-model and cross-section choices into nulls (it was found to be the decisive ingredient in reproducing JUNO's solar-parameter result to 0.06σ). It has no analogue for projections, where the measured flux with its covariance takes its place.

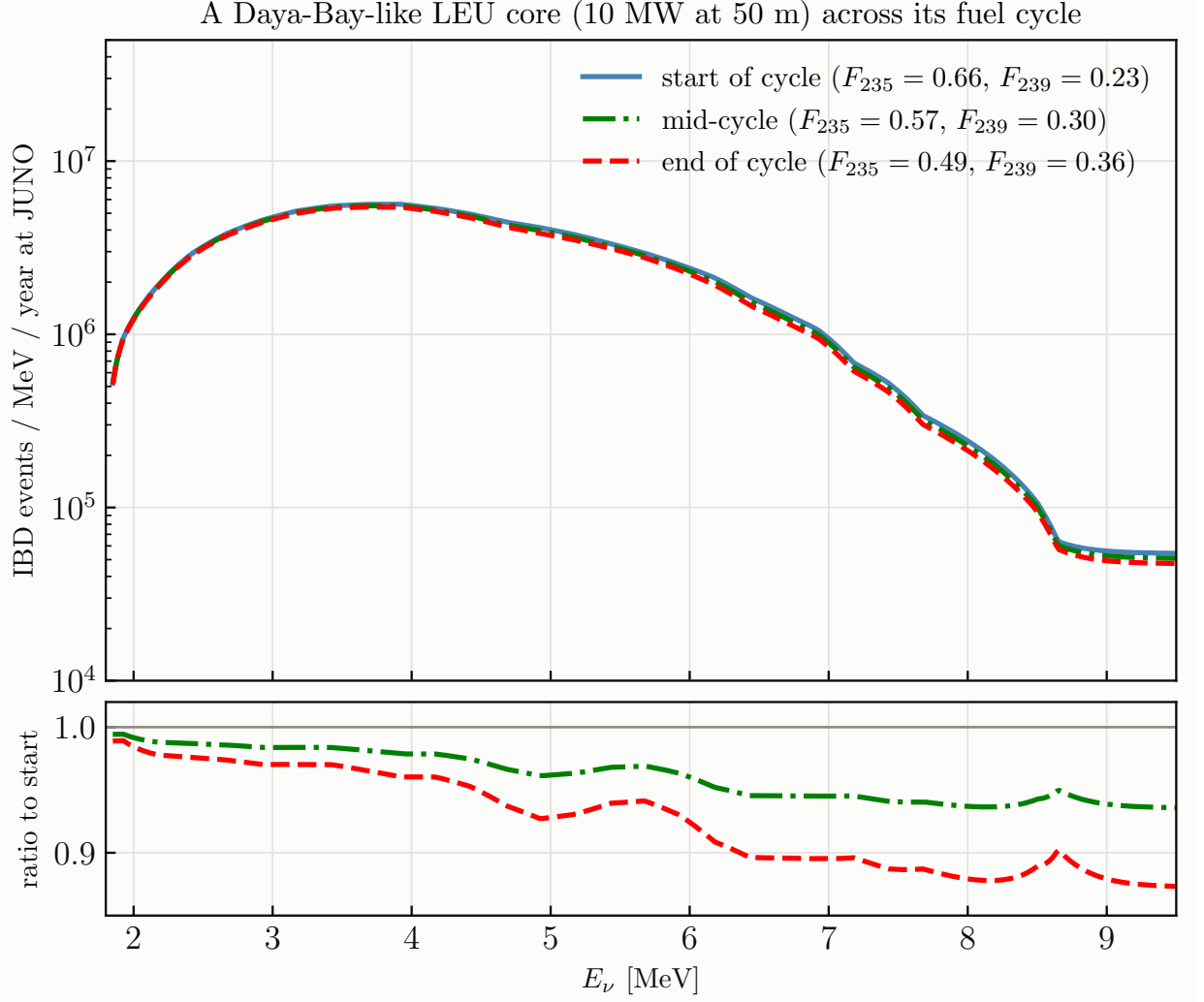


FIG. 16: Same for a commercial-LEU core: the Daya Bay measured fission fractions [2] at the start, middle and end of their 20 burnup groups, at the same 10 MW and 50 m so the two figures are directly comparable.

Appendix B: Systematics of the standard JUNO analysis (NuFit procedure)

The repository’s standard far-reactor analysis reproduces the NuFit 6.1 treatment of the JUNO 59.1-day data [7], and its nuisance set is what the near-reactor studies inherit for the far side and for the detector response. The test statistic is

$$\chi^2 = \sum_i \frac{(N_i^{\text{obs}} - N_i^{\text{pred}}(\boldsymbol{\xi}))^2}{\sigma_{\text{CNP},i}^2} + \sum_a \left(\frac{\xi_a}{\sigma_a} \right)^2, \quad \frac{3}{\sigma_{\text{CNP},i}^2} = \frac{1}{N_i^{\text{obs}}} + \frac{2}{N_i^{\text{pred}}}, \quad (\text{B1})$$

the combined Neyman–Pearson form used by JUNO (NuFit’s own default is the Poisson Baker–Cousins form $2 \sum_i [P_i - O_i + O_i \ln(O_i/P_i)]$; the two agree to 10^{-4} in $\sin^2 \theta_{12}$). The prediction

pull	meaning	prior σ
ξ_{norm}	signal normalisation	$2.4\% = 1.8\% \text{ (Tab. 2 rates)} \oplus 1.6\% \text{ (selection efficiency)}$
ξ_{scl}	energy scale	0.5%
ξ_{bias}	energy bias (offset in F_{nl} units)	0.5%
ξ_{res}	resolution	5%
ξ_{LiHe}	$^9\text{Li}/^8\text{He}$ normalisation	33%
ξ_{sh}	$^9\text{Li}/^8\text{He}$ linear-in- E shape (20% at 1 MeV)	0.20
ξ_{geo}	geoneutrino normalisation	42%
ξ_{world}	world-reactor normalisation	10%
ξ_{BiPo}	^{214}Bi – ^{214}Po normalisation	56%
ξ_{other}	other backgrounds	100%
$\xi_k, k = 1..25$	Daya Bay flux-shape modes (Section A)	1 each
r_{BG}	overall background rescaling (all but geoneutrinos)	fixed, 1.00 (NuFit cnf 2: 1.15)
$r_{\text{nl}}, r_{\text{res}}$	non-linearity / resolution rescalings	fixed, 1.0 (cnf 3: 1.024; cnf 5: 1.3)

TABLE III: Nuisance parameters of the standard analysis. The 2.4% normalisation is the NuFit “note added” budget; their published Tab. 1 configurations use 1.8%. External inputs:

$$\sin^2 \theta_{13} = 0.02175, \Delta m_{ee}^2 = 2.466 \times 10^{-3} \text{ eV}^2 \text{ (Daya Bay)}.$$

is

$$N_i^{\text{pred}} = (1 + \xi_{\text{norm}}) k_i \mathcal{N} \sum_{\text{cores } c} \left[R(\xi_{\text{scl}}, \xi_{\text{bias}}, \xi_{\text{res}}) S(E; \{\xi_k\}) \frac{P_{ee}(E, L_c)}{4\pi L_c^2} \right]_i + \sum_b B_{b,i} (1 + \xi_b) (1 + \delta_{b,\text{LiHe}} \xi_{\text{sh}} E_i), \quad (\text{B2})$$

with the nine signal cores (Yangjiang, Taishan and the effective Daya Bay core at 215 km), matter effects at 2.55 g cm^{-3} , and the response

$$\tilde{E}_{\text{pr}} = E_{\text{pr}} r_{\text{nl}} [(1 + \xi_{\text{scl}}) F_{\text{nl}}(E_{\text{pr}}) + \xi_{\text{bias}}], \quad \sigma(\tilde{E}) = (1 + \xi_{\text{res}}) r_{\text{res}} \sqrt{a^2 \tilde{E} + b^2 \tilde{E}^2}, \quad (\text{B3})$$

F_{nl} being JUNO’s released positron non-linearity and $a = 3.3\%$, $b = 1.0\%$. Table III lists every pull and its prior in the standard configuration.

a. Profiling. The prediction is affine in the three energy pulls (through the response matrix) and the 25 flux pulls (through the spectrum) at fixed oscillation parameters, so a linear basis is built once per grid point and the 28 pulls plus the background nuisances are profiled analytically; the two oscillation parameters are minimised numerically. In the Asimov projections of Sections II and III the same nuisances become the joint modes of $C = \text{diag}(\mu) + \sum_k v_k v_k^T$ and no minimisation is performed at all.

b. What the reproduction established. With this treatment the repository reproduces NuFit’s Tab. 2 relative fluxes exactly, their best fit to 0.10σ , and their Δm_{ee}^2 profile; with the note-added budget it lands on JUNO’s official $(\sin^2 \theta_{12}, \Delta m_{21}^2)$ at 0.06σ with every nuisance well inside its prior. The decisive ingredients, in order, were including the Daya Bay complex in the signal (its $\Delta_{21} \approx 2\pi$ position fills the solar dip), the bin-per-bin rescaling of Eq. (A4), and the 2.4% normalisation. The residual difference in the ellipse tilt corresponds to about 3% of effective rate freedom, that is, only $\sqrt{3^2 - 2.4^2} \approx 1.8\%$ beyond the documented budget, and it must be coherent in energy (candidates: the delivered-power history, the absolute Daya Bay flux anchor, the target proton count, and off-equilibrium/spent-fuel corrections).

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